Supplementary Material

A Droop Quota DAE Model for *Mycobacterium smegmatis* Batch Cultivation Under Nutrient-Replete and Nitrogen-Limited Conditions

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Conventions. *Nutrient-replete* and *ammonium-deficient* denote the two culture conditions. The datasets of the experimental workbook are called by their names: PE = calibration dataset (0–40 h), V = validation dataset (0–99.5 h). The *baseline model* is that of Díaz-Figueroa *et al.* [1] (Sect. S1.1) and the *extended model* is the one of the main text (Sect. S1.2). The *initial set* is the extended model with the values of the series files (column "Initial" of Table S1), and the *selected set* (M5) is the one chosen in Sect. S3.7. NRMSE (per variable and global) is defined exactly as in the main text, and AIC, AICc and BIC in Sect. S4.1 ($n = 40$ observations per dataset). Concentrations are in g/L (nitrogen in g N/L) and time in hours. Terms for the calibration: a *run* is one PSO optimization from one random seed and returns one *solution* (parameter set); a *series* is a set of runs (ten planned) of one *calibration subset* (the parameters that are calibrated, the others fixed) that differ only in the seed; *rounds* 1–5 are the successive steps of the exploratory stage, in which the calibration subsets and bounds were changed by hand, and the *final stage* uses bounds fixed in advance; M4 to M7 are the calibration subsets of the final-stage ladder, numbered by their calibrated parameters. Numbers in the text come from the same result tables as the main text and are generated by scripts, so they agree with it.

S1. Model specification

A. Baseline model

The baseline model has the states (X, C, N, CO_2, O_2) and the algebraic variable pH:

$$\dot{X} = (\mu_D - k_d)X, \quad \dot{C} = -\frac{\mu_D}{Y_{X/C}}X, \quad \dot{N} = -\frac{\mu_D}{Y_{X/N}}X, \tag{S1}$$

$$\dot{CO_2} = \frac{\mu_D}{Y_{X/CO_2}}X - k_{a7}\frac{CO_2}{10^{-pH}/k_{a7} + 1}, \quad \dot{O}_2 = k_L a(O_{sat} - O_2) - \frac{\mu_D}{Y_{X/O_2}}X, \tag{S2}$$

$$\mu_D = \mu_{\max}\big(1 - e^{-t/t_{lag}}\big)\frac{C}{C + k_C}\frac{N}{N + k_N}\frac{O_2}{O_2 + k_O}\Big(1 - \frac{X}{X_{\max}}\Big)I^D_{pH}, \tag{S3}$$

$$I^D_{pH} = \exp\Big[\Big(I_{val}\frac{pH - pH_{UL}}{pH_{UL} - pH_{LL}}\Big)^2\Big], \tag{S4}$$

with the same constraint as the extended model but without the ammonium term (Sect. S1.3). As implemented in the code of [1], which is kept for a faithful comparison, I^D_{pH} equals one at pH_{UL} and grows on both sides, so it is not a bounded inhibition factor. The baseline has 14 parameters (Table S1); three of them ($Y_{X/N}$, $X_{\max}$, $\mu_{\max}$) were calibrated in [1] on the nutrient-replete calibration dataset and the rest are fixed. The main text uses $Y_{X/N} = 10.5$ g/g, the value in the text of [1]; the package data file holds 10.4493. With either value the global NRMSE of the baseline model on the four datasets (PE_Normal, V_Normal, PE_Amm_def, V_Amm_def_rep) differs by at most 0.06 percentage points (18.5, 12.3, 47.4 and 41.2% with 10.5; 18.6, 12.3, 47.4 and 41.3% with 10.4493). The values stored for the baseline before its calibration are $Y_{X/N} = 21.575$ g/g (the value of Apiyo *et al.* [2], their Table 3), $X_{\max} = 1.446$ g/L and $\mu_{\max} = 0.15$ h^{-1}; the values after calibration are 10.5, 1.1537 and 0.1335. For the calibration settings of the baseline, including its bounds, see [1].

B. Extended model

The state vector is $x = (X, C, N, CO_2, O_2, q)$ with units g/L, g/L, g N/L, g/L, g/L and g N/g X, plus the algebraic variable pH. With t in hours, the kinetic factors are

$$f_{lag} = 1 - e^{-t/t_{lag}}, \qquad f_C = \frac{C}{C + k_C}, \qquad f_O = \frac{O_2}{O_2 + k_O},$$

$$f_{cap} = \max\Big(1 - \frac{X}{X_{\max}}, 0\Big)^{n_{cap}}, \qquad I_{pH} = \exp\Big[-\Big(\frac{pH - pH_{opt}}{pH_w}\Big)^2\Big],$$

$$f_{Nq} = \max\Big(1 - \frac{q_{\min}}{q}, 0\Big), \qquad f_{fill} = \max\Big(\frac{q_{\max} - q}{q_{\max} - q_{\min}}, 0\Big), \tag{S5}$$

the specific growth, uptake and maintenance rates are

$$\mu = \mu_{\max}\, f_{lag}\, f_C\, f_O \underbrace{\max\Big(1 - \frac{X}{X_{\max}}, 0\Big)^{n_{cap}}}_{f_{cap}} I_{pH}\, f_{Nq}, \qquad \rho = \rho_{N,\max}\frac{N}{N + k_{Nup}} f_{fill},$$
$$m(X) = m_C \max\Big(1 - \frac{X}{X_{\max}}, 0\Big)^{m_{cap}}, \tag{S6}$$

and the balances are

$$\dot{X} = (\mu - k_d)X, \qquad \dot{C} = -\Big(\frac{\mu}{Y_{X/C}} + m(X)\Big)X, \tag{S7}$$
$$\dot{N} = -\rho X + f_{Nrec}\, k_d\, q\, X, \qquad \dot{q} = \rho - \mu q, \tag{S8}$$
$$\dot{CO_2} = \Big(\frac{\mu}{Y_{X/CO_2}} + m(X)\, Y^m_{CO_2}\Big)X - k_{a7}\frac{CO_2}{10^{-pH}/k_{a7} + 1}, \tag{S9}$$
$$\dot{O}_2 = k_L a\,(O_{sat} - O_2) - \Big(\frac{\mu}{Y_{X/O_2}} + m(X)\, Y^m_{O_2}\Big)X, \tag{S10}$$
$$0 = \text{electroneutrality constraint (Sect. S1.3).} \tag{S11}$$

Structure and sources. The Droop quota q and the factor f_{Nq} follow the Droop model [3] in the form used for nitrogen by Bougaran *et al.* [4]; the filling factor f_{fill} is a feedback inhibition of uptake of the same type as that used for *M. smegmatis* by Apiyo *et al.* [2] (Sect. S1.5). The Droop mechanism comes from phytoplankton ecology and is used here as a structural transfer; no source gives the parameters of either mechanism for *M. smegmatis*. Setting $q_{\max} \to \infty$ removes the feedback inhibition of uptake, and $n_{cap} = m_{cap} = 1$ recovers the logistic capacity term and a linear decay of maintenance.

Han–Levenspiel-type self-inhibition. The factor f_{cap} is the cell-inhibition term of Han and Levenspiel [5] with critical biomass $X_{\max}$ and exponent n_{cap}. Their original model also scales the Monod constants with this term; that scaling is not used here, hence the name "type". The maximum with zero keeps the base non-negative, which matters for non-integer exponents. The factor equals one at $X = 0$ for every n_{cap} and reduces to the logistic term of the baseline model at $n_{cap} = 1$. For $n_{cap} < 1$ the derivative of f_{cap} with respect to X is unbounded at $X = X_{\max}$, which makes the integration harder close to the ceiling. This was tested in two ways.

(i) Grid test. The integrator (CasADi/IDAS) was run for $n_{cap} \in \{0.2, 0.3, 0.4, 0.5, 0.8, 1, 2, 3\}$ and $m_{cap} \in \{0.5, 1, 1.5, 3\}$ (two parameter sets $\times$ 32 combinations = 64 integrations over 0–100 h with the ammonium-deficient validation initial conditions): a stress set with fast growth that drives X to $X_{\max}$ ($\mu_{\max} = 0.30$ h^{-1}, $t_{lag} = 5$ h, $X_{\max} = 1$ g/L, the other parameters as in the column "Initial" of Table S1), and the selected set with n_{cap} and m_{cap} replaced. Every integration with $n_{cap} \geq 0.4$ finished with finite states. The integration failed (the solver raised an error) at $n_{cap} = 0.2$ for every m_{cap} in both sets and, in the stress set, at $n_{cap} = 0.3$ with $m_{cap} = 1$ and 3. The selected value $n_{cap} = 0.3793$ lies in the region where failures are possible for other parameter sets, although it integrates without failure at the selected values of the other parameters.

(ii) Failed objective evaluations. Each PSO run evaluates the objective 10 050 times. Runs of rounds 1–5 that did not calibrate n_{cap} had no failed evaluation (120 runs). Among the runs of rounds 1–5 that calibrated n_{cap}, the numbers of runs with failed evaluations (mean and maximum number of failures per run) were 0 of 372 for a lower bound of 0.5 or 0.4, 1 of 5 (maximum 1) for 0.35, 46 of 50 (mean 6.8, maximum 51) for 0.3 and 5 of 5 (mean 32.6, maximum 40) for 0.1. The ten runs of the selected series had between 3 and 8 failed evaluations each (5 for the selected run). In the final stage (lower bound 0.3) 117 of the 124 runs that calibrated n_{cap} had failed evaluations (mean 7.8, maximum 80), and none of the ten runs without n_{cap}. A failed evaluation is penalized by the optimizer and is not part of the reported solution.

Biomass-dependent maintenance. A constant Pirt-type maintenance coefficient [6] keeps consuming glycerol at high cell density. $m(X)$ decays with biomass and reduces to a linear decay at $m_{cap} = 1$ and to the constant coefficient m_C at $m_{cap} = 0$. $m(X)$ is not multiplied by the glycerol saturation term f_C, so maintenance continues at any glycerol concentration, and the model does not enforce $C \geq 0$; the minimum simulated glycerol of the selected set is 2.85 g/L in the ammonium-deficient datasets and 2.75 g/L in the nutrient-replete ones (lowest measured values 3.16 and 3.21 g/L, Tables S6–S9; the simulation therefore consumes more glycerol than was measured, which is the largest error of the ammonium-deficient validation, Table S16). The decay with X is phenomenological, in the spirit of the growth-dependent maintenance adaptations discussed in [7], and not a validated mechanism for *M. smegmatis*. Its effect is tested in Sect. S6.

The model is an index-1 DAE (the constraint can be solved for pH at every state, see Sect. S1.3) with six differential states, one algebraic variable and 21 parameters (Table S1); 5 of them are calibrated.

TABLE S1: Parameters of the baseline and extended models. *Baseline*: value in the baseline model (calibrated ones in **bold**). *Initial*: value of the series (`base_parameters` of the series files; $pH_w = 7$ in rounds 1–5 and 10 in the final stage, the other values are the same), from which the parameters that are not calibrated are never changed. *Selected*: M5. Status C: calibrated, F: fixed. *Widest range* and *final bounds*: bounds of the calibrated parameters in the 59 series (rounds 1–5 and final stage) and in the series of the selected set (`seventh_run_0`). Source: B: value of the baseline model [1]; A: assumed, no numerical source; Q: initial quota, see Sect. S1.4.

Parameter	Unit	Baseline	Initial	Selected	Status	Widest range	Final bounds	Source
k_C	g/L	0.08	0.08	0.08	F	—	—	B
k_O	g/L	0.001	0.001	0.001	F	0.0001–0.1	—	B
k_d	1/h	0.001	0.001	0.001	F	—	—	B
Y_{X/CO_2}	g X/g CO_2	0.35	0.35	**0.3591**	C	0.05–5	0.05–1	B
Y_{X/O_2}	g X/g O_2	1.352	1.352	1.352	F	—	—	B
$Y_{X/C}$	g X/g C	0.484	0.484	**0.3391**	C	0.05–5	0.05–1	B
$X_{\max}$	g/L	**1.154** (C)	1.15	**1.025**	C	0.8–2.5	0.8–2.5	B
$\mu_{\max}$	1/h	**0.1335** (C)	0.1335	**0.118**	C	0.05–1	0.05–1	B
t_{lag}	h	7	7	7	F	1–60	—	B
m_C	g C/(g X·h)	–	0.008	0.008	F	0–0.5	—	A
f_{Nrec}	–	–	0.5	0.5	F	—	—	A
pH_{opt}	–	–	7	7	F	2–12	—	A
pH_w	–	–	10	10	F	1–10	—	A
$Y^m_{CO_2}$	g CO_2/g C	–	0.05	0.05	F	—	—	A
$Y^m_{O_2}$	g O_2/g C	–	1	1	F	—	—	A
$\rho_{N,\max}$	g N/(g X·h)	–	0.5	0.5	F	0.01–0.5	—	A
k_{Nup}	g N/L	–	0.0001	0.0001	F	—	—	A
$q_{\min}$	g N/g X	–	0.0001	0.0001	F	0–0.05	—	A
$q_{\max}$	g N/g X	–	0.096	0.096	F	0.096–0.16	—	Q
n_{cap}	–	–	0.8	**0.3793**	C	0.1–5	0.3–5	A
m_{cap}	–	–	1.5	1.5	F	0–15	—	A
k_N	g N/L	0.1	—	—	—	—	—	B
$Y_{X/N}$	g X/g N	**10.5** (C)	—	—	—	—	—	B
pH_{LL}, pH_{UL}, I_{val}	—	4.6, 7.4, 3	—	—	—	—	—	B

C. Electroneutrality constraint

With $[H^+] = 10^{-pH}$, $K_{ai} = 10^{-pK_{ai}}$ (Table S2) and $K_w = K_{a9}$, the terms of the constraint as implemented in the package are

$$[HCO_3^-] = \frac{CO_2}{1 + [H^+]/K_{a7}}, \qquad P_B = \frac{[\mathrm{KH_2PO_4}]}{1 + [H^+]/K_{a1}}, \qquad [OH^-] = \frac{K_w}{[H^+]}, \tag{S12}$$

$$[Cit^{3-}] = \frac{[\mathrm{C_6H_8O_7}]}{1 + [H^+]/K_{a4} + [H^+]^2/(K_{a3}K_{a4}) + [H^+]^3/(K_{a2}K_{a3}K_{a4})}, \tag{S13}$$

$$[HCit^{2-}] = \frac{[H^+]}{K_{a4}}[Cit^{3-}], \qquad [H_2Cit^-] = \frac{[H^+]}{K_{a3}}[HCit^{2-}], \qquad [NH_4^+] = \frac{N}{M_N}\frac{[H^+]}{[H^+] + K_{a5}}, \tag{S14}$$

$$0 = [OH^-] + [HCO_3^-] + P_B + 3[Cit^{3-}] + 2[HCit^{2-}] + [H_2Cit^-] - \mathrm{pH}_{alk} - [H^+] - [NH_4^+]. \tag{S15}$$

The second dissociation of carbonic acid is omitted. The baseline model uses the same expression without $[NH_4^+]$. P_B is the concentration of the base form of phosphate (HPO_4^{2-}) and enters with unit weight. Every term of the constraint other than pH_{alk} and $-[H^+]$ increases with pH, so the left-hand side is strictly increasing in pH and has one root in the physical range: this is why the DAE has index 1.

D. Provenance of parameters and constants

The acid–base constants, the medium concentrations that enter the electroneutrality constraint, the oxygen-transfer constants and the yield coefficients shared with the baseline model were traced to the thesis of Apiyo [8], the source of the experimental data and of the original model. Secondary references are as listed in [8] and were not independently verified. Two points follow.

Charge-balance constants. The pK_a values come from Table 3-15 of [8]; phosphate and citrate are the concentrations of the modified Sauton medium (Table 3-2 of [8]), entered as g/L numbers; and pH_{alk} is the lumped constant-ion term I_{cst} of [8], fixed here at 7.20 for every dataset (the value used in [8] for the replicate replete culture). The initial CO_2 of each dataset is back-calculated so that the constraint reproduces the measured initial pH (Table S5). No temperature correction of the pK_a values is applied.

Parameters without a source. The values of the twelve parameters that the baseline model does not have (m_C, m_{cap}, n_{cap}, f_{Nrec}, pH_{opt}, pH_w, $Y^m_{CO_2}$, $Y^m_{O_2}$, $\rho_{N,\max}$, k_{Nup}, $q_{\min}$ and $q_{\max}$) have no numerical source in [8]; they are assumed values (Table S1, source A), set by the authors on physical and biological grounds rather than measured for *M. smegmatis*, and they were not calibrated. Table S3 gives, for each of them, the reason that supports its magnitude and the range over which it can be

TABLE S2: Constants fixed and identical for both models. Values as read from the package data files; pK_{a6} and pK_{a8} are present in the files but do not enter Eq. (S15). Subscripts follow the package.

Constant	Value	Meaning / unit
pK_{a1}	6.86	$H_2PO_4^-$/HPO_4^{2-}
pK_{a2}	3.13	citrate, H_3Cit/H_2Cit^-
pK_{a3}	4.76	citrate, $H_2Cit^-/HCit^{2-}$
pK_{a4}	6.40	citrate, $HCit^{2-}/Cit^{3-}$
pK_{a5}	9.25	NH_4^+/NH_3
pK_{a7}	6.35	CO_2/HCO_3^-
pK_{a9}	14	water (K_w)
pK_{a6}, pK_{a8}	14.15, 10.33	listed in the package; not used in the charge balance of these models
$[KH_2PO_4]$	2.18	phosphate, entered numerically as g/L
$[C_6H_8O_7]$	2	citrate, entered numerically as g/L
pH_{alk}	7.20	lumped constant-ion term (Sect. S1.4)
M_N	14.007	g/mol
k_La	86.26	1/h
O_{sat}	0.0072673	g/L (7.2673 mg/L)

changed: each parameter was changed one at a time and the five calibrated parameters were re-optimised (local least squares started at the selected set; script `sens_assumed.py` in the repository), and the range is the one in which the global NRMSE of every dataset stays within 0.5 points of the selected set. A wide range means that the data carry no information on the parameter, not that the value is correct, and within the ranges some calibrated parameters shift to keep the fit (last column of the table). The initial $n_{cap} = 0.8$ is only the starting value of the series; local re-optimisations of the five calibrated parameters started at $n_{cap} = 0.8$, 2 and 5 all ended at 0.379. The exponents n_{cap} and m_{cap} are introduced in this work; no value for *M. smegmatis* is available, and values reported for other systems in [5] are not transferred. The initial quota $q(0) = 0.096$ g N/g X, equal to $q_{\max}$ in the selected set, is the nitrogen mass fraction implied by the biomass formula of Sample 1 of [8] and was not measured in these cultures; with $q(0) = q_{\max}$ the uptake feedback is closed at the start of every culture, so the quota can only fall below $q_{\max}$ through growth. The series files also contain a run in which $q_{\max}$ was calibrated with $q(0) = q_{\max}$ (Sect. S6).

TABLE S3: Assumed parameters: value, reason for its magnitude and range over which it can be changed. *Range*: values tested one at a time, with the five calibrated parameters re-optimised (local least squares started at the selected set), for which the global NRMSE of each of the four datasets stays within 0.5 points of the selected set. A wide range means that the data carry no information on the parameter; it does not show that the value is correct. *Shift*: largest change of a calibrated parameter within the range (the fit is kept by compensation).

Parameter	Value	Unit	Reason for the magnitude	Range	Shift
m_C	0.008	g C/(g X·h)	Small: maintenance is only 1% of the glycerol drawn. Larger values are paid for by a larger $Y_{X/C}$.	0–0.1	$Y_{X/C}$ +22%
m_{cap}	1.5	–	Exponent of the decline of maintenance as $X \to X_{\max}$. The calibration data do not identify it (0–15 fit equally well); $m_{cap} = 0$ worsens the validation by 3 points.	0.5–15	$Y_{X/C}$ −2%
$\rho_{N,\max}$	0.5	g N/(g X·h)	Far above the peak nitrogen demand ($\mu q \approx 0.007$ g N/(g X·h)), so uptake never limits the quota. Calibrated in the final stage (M6, Sect. S3.6) it is 0.08658 (95% half-width 48%), inside the range, with no gain in validation.	0.05–2	n_{cap} +45%
k_{Nup}	10^{-4}	g N/L	Far below the ammonium of the medium (0.03–0.04 g N/L ammonium-deficient, about 1 g N/L replete, Table S5): uptake is saturated until the medium is nearly empty.	7×10^{-7}–0.1	n_{cap} +43%
$q_{\min}$	10^{-4}	g N/g X	Effectively zero: the Droop factor is about 1 (no limitation). Whether a larger value is compatible with the data is not settled (Sect. S6); the range ends where a nutrient-replete simulation of the re-optimised set fails to integrate, not where the fit worsens.	0–0.032	$\mu_{\max}$ +40%
$q_{\max}$	0.096	g N/g X	Equal to the initial quota $q(0)$, the nitrogen fraction of the biomass formula (Sect. S1.4). The data confine it to a narrow range; calibrated together with $\rho_{N,\max}$ (M7) it is 0.09817.	0.0955–0.1	n_{cap} −14%
f_{Nrec}	0.5	–	Midpoint of its possible range 0–1; nothing is known on the share of the quota released by dying cells.	0–1	n_{cap} +2%
$Y^m_{CO_2}$	0.05	g CO_2/g C	Not constrained: CO_2 was not measured and reaches the measured variables only through pH. Complete oxidation of glycerol (1.43 g CO_2/g with $Y^m_{O_2} = 1.22$) changes the NRMSE by at most 0.23 points.	0–3	Y_{X/CO_2} +6%
$Y^m_{O_2}$	1	g O_2/g C	Not constrained: O_2 was not measured; after the first 0.5 h it stays at about 95% of saturation, so the O_2 factor is nearly constant (0.87).	0–4	none ($< 1\%$)
pH_{opt}	7	–	Neutral, within the initial pH of the cultures (6.9–7.2, Table S5), tested at $pH_w = 10$.	5–9	$\mu_{\max}$ +5%
pH_w	10	–	Wide: the factor stays ≥ 0.99 over the simulated pH 6.4–7.2 (0.996 at 10; 0.993 at 7, the value of the first five rounds, with the same fit to within 0.01 points), i.e. pH inhibition is off in practice; the cultures span only about 0.8 pH units.	2–14	n_{cap} −10%

E. Relation to the internal nitrogen storage of Apiyo et al.

Apiyo *et al.* [2] modelled an internal nitrogen storage N_{int} (g NH_3/g biomass) with the uptake rate $J_N = k_L \frac{N}{N+k_J}\left(1-\frac{N_{int}}{N^*}\right)$ (their Eq. 10), a growth rate with the Monod factor $N_{int}/(N_{int}+k_N)$ (their Eq. 3), the pool balance $\mathrm{d}(N_{int}X)/\mathrm{d}t = J_N - \mu X/Y_{X/N}$ (their Eq. 11) and ammonium in the charge balance (their Eq. 14). The extended model keeps the same type of uptake feedback (the factor $f_{fill} = (q_{\max}-q)/(q_{\max}-q_{\min})$ of Eq. (S5) has the form of $1 - N_{int}/N^*$ when $q_{\min} = 0$) and differs in four points:

- *Growth.* Growth is limited by the Droop factor $f_{Nq} = \max(1 - q_{\min}/q, 0)$ of the total cell quota, not by a Monod factor of the storage.
- *Pool balance.* Here q is the total nitrogen per unit of biomass and growth only dilutes it (Eq. (S8)); in [2] growth withdraws nitrogen from the pool at the fixed rate $1/Y_{X/N}$ per unit of biomass formed.
- *Death.* Dying cells return a fraction f_{Nrec} of their quota to the medium; in [2] $\mathrm{d}N/\mathrm{d}t = -J_N$, with no return.
- *Lag and acid.* The lag of [2] is a state μ_{int} with a time constant τ (their Eq. 2) and they include an unspecified acid A produced in proportion to growth (their Eq. 9); here the lag is the factor $1 - e^{-t/t_{lag}}$ and the pH follows from the phosphate, citrate, ammonium and carbonate terms only.

The parameter values are not comparable: $N^* = 0.0306$ g NH_3/g (their Table 4) is a storage capacity, whereas $q_{\max} = 0.096$ g N/g X here is a total quota.

S2. Experimental data

The data were provided by Apiyo *et al.* [2] (batch cultures of *M. smegmatis* on glycerol) and are stored in four datasets of the experimental workbook (Table S4). The initial ammonium nitrogen is about 1 g N/L in the nutrient-replete cultures and 0.03 g N/L in the ammonium-deficient ones. Each dataset lists the mean and the standard error (SE) of X, C, N and pH at ten sampling times (Fig. S1, Tables S6–S9). The workbook does not store the number of replicates behind each mean, and some SE entries are zero (for example X at 19.5 h in PE_Amm_def); we do not interpret them. Entries stored as 6.28×10^{-16} (the SE of the initial pH of PE_Amm_def) are numerical zeros and are printed as 0. CO_2 and O_2 were not measured. The workbook has further datasets (PE_pH_55, PE_pH_46, PE_Glyc_def, V_Glyc_def, V_Amm_Glyc_def and V_Amm_def) that are outside the scope of this paper. V_Amm_def_rep, and not V_Amm_def, is the ammonium-deficient validation dataset because V_Amm_def is a spiking experiment and was not used in any experiment of this paper, whereas V_Amm_def_rep is the culture without spiking.

TABLE S4: Datasets used in this work and their role for each model.

Dataset	Condition	Samples	Window (h)	Baseline model	Extended model
PE_Normal	nutrient-replete	10	0–40	calibration [1]	test, no refit
V_Normal	nutrient-replete	10	0–99.5	validation	validation, no refit
PE_Amm_def	ammonium-deficient	10	0–40	test, no refit	calibration (PSO objective and selection)
V_Amm_def_rep	ammonium-deficient	10	0–99.5	validation	validation

TABLE S5: Initial states of each dataset. X, C, N and pH are the first sample of the dataset; $CO_2(0)$ is the value stored in the package, which reproduces the measured initial pH through the charge balance with the constants of Table S2 (checked: the nutrient-replete datasets with the ammonium term and the ammonium-deficient ones without it; with the ammonium term the latter would start at pH 6.901 and 7.098, 0.001 units above the measured values); $O_2(0)$ is O_{sat} in the calibration datasets and 10^{-4} g/L in the validation datasets; $q(0) = 0.096$ g N/g X in all datasets (assumed, not measured).

Dataset	$X(0)$	$C(0)$	$N(0)$	pH(0)	$CO_2(0)$	$O_2(0)$
PE_Normal	0.2285	5.3891	0.9508	7.157	0.1377	0.007267
PE_Amm_def	0.2241	6.0568	0.0363	6.900	0.7005	0.007267
V_Normal	0.2230	5.9224	1.0266	7.183	0.0911	0.0001
V_Amm_def_rep	0.2219	5.5114	0.0317	7.097	0.1860	0.0001

Weights. The two calibration datasets also store a weight per observation. For each of the 74 entries (of 80) with a non-zero SE it equals $1/(3\,\mathrm{SE}^2)$, and it is 1 for the six entries with zero SE. The weights are not used in the PSO objective (Sect. S3) nor in the FIM behind the reported t-values (Sect. S7.2), where the error variance of each variable comes from the calibration residuals; the package keeps them only for an optional replicate-weight mode of the FIM, which was not used for the reported results. The stored SE of N in V_Normal at 51.5 h (0.142 g N/L) is about eleven times the median SE of N in that dataset (0.0129 g N/L); it is plotted as stored.

Apparent biomass yield on nitrogen. From the first and last sample of each validation dataset, $\Delta X/\Delta N = (1.193 - 0.223)/(1.027 - 0.948) \approx 12$ g/g for V_Normal and $(1.116 - 0.222)/(0.0317 - 0.00007) \approx 28$ g/g for V_Amm_def_rep, a ratio of 2.3. This is arithmetic on the stored means and is independent of any model; it ignores sampling error (ten points per variable) and the nitrogen content of dead biomass.

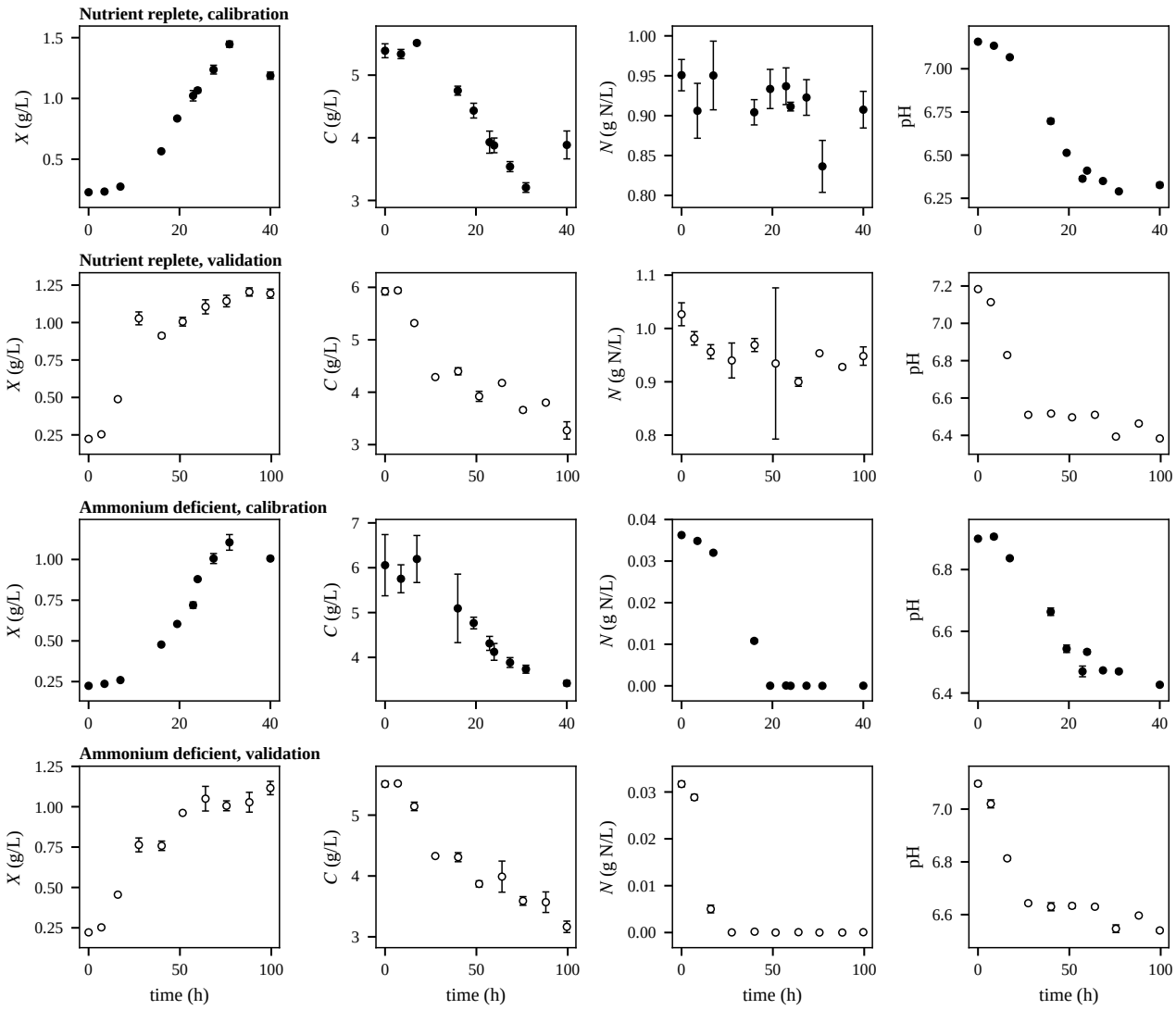


Fig. S1: Measured data (mean ± SE) of the four datasets. Filled symbols: calibration datasets (PE); open symbols: validation datasets (V).

TABLE S6: Nutrient-replete calibration dataset (PE_Normal): mean and standard error (SE) of X, C (g/L), N (g N/L) and pH.

t (h)	X	SE	C	SE	N	SE	pH	SE
0	0.2285	0.0011	5.389	0.111	0.9508	0.0196	7.157	0.00333
3.5	0.234	0	5.337	0.0734	0.9061	0.0344	7.133	0.00667
7	0.2748	0.0011	5.514	0.0291	0.9504	0.043	7.067	0.00333
16	0.5654	0.011	4.752	0.0719	0.9042	0.0158	6.697	0.0145
19.5	0.8352	0.00953	4.434	0.118	0.9335	0.0246	6.513	0.00667
23	1.022	0.043	3.93	0.177	0.9369	0.023	6.363	0.012
24	1.066	0.0191	3.879	0.118	0.9113	0.00545	6.41	0.00577
27.5	1.237	0.0361	3.539	0.0802	0.9228	0.0223	6.35	0.00577
31	1.446	0.0252	3.205	0.0767	0.8364	0.0326	6.29	0.00577
40	1.187	0.0291	3.886	0.223	0.9075	0.023	6.327	0.0133

S3. Calibration procedure and data usage

A. *Objective and settings*

PSO (`mealpy`, `OriginalPSO`, 50 particles, 200 epochs, $c_1 = c_2 = 1.5$, inertia 0.5; read from the series file of every one of the 59 series analysed here, which have identical settings; the final-stage series also share the range-scaled cost and the

TABLE S7: Nutrient-replete validation dataset (V_Normal): mean and SE.

t (h)	X	SE	C	SE	N	SE	pH	SE
0	0.223	0.0011	5.922	0.0674	1.027	0.0215	7.183	0.00667
7	0.2538	0.0033	5.94	0.044	0.9814	0.0126	7.113	0.00333
16	0.4884	0.00953	5.318	0.0301	0.9563	0.0132	6.83	0
27.5	1.028	0.043	4.288	0.0259	0.9399	0.0328	6.51	0
40	0.9122	0.0146	4.399	0.0675	0.9689	0.0122	6.517	0.00667
51.5	1.006	0.0291	3.921	0.0962	0.9343	0.142	6.497	0.00333
64	1.105	0.047	4.176	0.0327	0.8998	0.00821	6.51	0
75.5	1.143	0.0385	3.661	0.0406	0.9535	0.000721	6.393	0.00667
88	1.204	0.0275	3.801	0.0264	0.9278	0.00355	6.463	0.00882
99.5	1.193	0.0307	3.271	0.165	0.9482	0.0172	6.383	0.00333

TABLE S8: Ammonium-deficient calibration dataset (PE_Amm_def): mean and SE.

t (h)	X	SE	C	SE	N	SE	pH	SE
0	0.2241	0	6.057	0.683	0.03626	0.000118	6.9	0
3.5	0.2362	0.0011	5.753	0.312	0.03484	0.000433	6.907	0.00667
7	0.2593	0.0022	6.195	0.525	0.032	0.000401	6.837	0.00333
16	0.4773	0.00551	5.093	0.764	0.0108	0.000423	6.663	0.012
19.5	0.604	0	4.766	0.13	2.327e-05	2.33e-05	6.543	0.012
23	0.7196	0.0191	4.312	0.156	6.567e-05	3.3e-05	6.47	0.0173
24	0.8792	0.0055	4.123	0.187	0	0	6.533	0.00667
27.5	1.006	0.0307	3.885	0.11	2.207e-05	2.21e-05	6.473	0.00333
31	1.105	0.048	3.736	0.0884	0	0	6.47	0.00577
40	1.006	0.011	3.423	0.0577	2.45e-05	2.45e-05	6.427	0.00333

TABLE S9: Ammonium-deficient validation dataset (V_Amm_def_rep): mean and SE.

t (h)	X	SE	C	SE	N	SE	pH	SE
0	0.2219	0.0011	5.511	0.0446	0.03169	0.000588	7.097	0.00882
7	0.2527	0.00291	5.52	0.018	0.02886	0.000601	7.02	0.0153
16	0.4553	0	5.143	0.0693	0.00502	0.000815	6.813	0.00667
27.5	0.7636	0.043	4.328	0.0145	2.647e-05	2.65e-05	6.643	0.00333
40	0.7581	0.0291	4.309	0.0745	0.000183	0.000183	6.63	0.0153
51.5	0.9618	0.0055	3.87	0.0497	0	0	6.633	0.00882
64	1.05	0.0763	3.989	0.256	7.553e-05	7.55e-05	6.63	0.00577
75.5	1.006	0.0307	3.589	0.0712	0	0	6.547	0.0145
88	1.028	0.0613	3.569	0.169	0	0	6.597	0.00333
99.5	1.116	0.0416	3.164	0.0938	7.033e-05	7.03e-05	6.54	0.00577

ammonium-deficient calibration dataset) minimized the sum of squared residuals between the simulation and the ammonium-deficient calibration dataset,

$$J = \sum_{v\in\{X,C,N,pH\}} \sum_{i=1}^{10} \left(\frac{y_{v,i} - \hat{y}_{v,i}}{R_v} \right)^2, \tag{S16}$$

with R_v the range of variable v in the dataset (`cost_scaling='range'` in every series; the package default is the unscaled sum in native units, in which glycerol, the variable with the largest numerical range, dominates the objective), and with the initial states of Table S5 (calibration dataset, $O_2(0) = O_{sat}$). With range scaling, $J = n\,\mathrm{NRMSE}^2_{glob}$ on the calibration dataset, so that minimizing the objective and minimizing the global NRMSE of the main text are the same problem. No validation dataset and no nutrient-replete dataset entered the objective. Each run of a series uses its own random seed.

B. Sequential estimation

The calibration subset and the bounds of the extended model were first reached in five rounds of series (Table S10; each series is in Table S15), in which the bounds of each series were narrowed by hand from the results of the previous one; the 552 runs of 42 series of these rounds are described below. A final stage then repeated the selection with bounds and starting values fixed in advance for every series (134 runs, 17 series; Sect. S3.6); the reported set comes from it. In all, 686 runs in 59 series were analysed. The results folder also holds two series, `base_original` and `base_original1`, that contain no calibration run: each only evaluates the starting parameter set of the extended model (the shared parameters take the values of the baseline of [1], the rest are assumed values) before any calibration, and both give the same result. They are the source of the initial set in the tables and are not counted as runs.

Round 1 (240 runs, 3 series). Wide bounds on 10 to 12 free parameters at once (`wide_1`, 65 runs; `wide_2`, 115 runs) or on each of 12 parameters alone (`wide_3`, 60 runs: five runs per parameter, the other parameters at their initial values).

Round 2 (45 runs, 9 series). Pools of 11 down to 6 parameters, smaller than in round 1.

Round 3 (30 runs, 6 series). Pools of 4 to 6 parameters around $\{\mu_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}\}$, with $X_{\max}$, pH_{opt}, $q_{\min}$ and $q_{\max}$ added in different series; $n_{cap} \in [0.5, 5]$.

Round 4 (117 runs, 11 series). The same four parameters with $X_{\max}$ (4 to 6 parameters per run; one series of 10 to 11) and, in one series each, t_{lag}, m_C, $q_{\max}$ or pH_{opt} added. The lower bound of n_{cap} was 0.5 in six series and 0.3 in five (`fourth_run_6` to `_10`), see Sect. S3.4.

Round 5 (110 runs, 11 series). Pools of 4 to 7 parameters around $\{\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}\}$: that pool with lower bounds of n_{cap} of 0.5, 0.4, 0.35 and 0.3 (`fifth_run_1`, `_4`, `_5`, `_3`); the pool without n_{cap} (`fifth_run_2`); and variants with t_{lag} and/or $q_{\max}$ added to it or replacing n_{cap} (`fifth_run_6` to `_11`). The selected set of the five rounds came from `fifth_run_3` (ten runs, lower bound 0.3); it is called M5_fifth in Table S20.

Additional series (10 runs, 2 series). The four-parameter pool without $X_{\max}$, with initial values $n_{cap} = m_{cap} = 1$ (`dif_start_run_1`) and with $n_{cap} \in [0.1, 5]$ (`ncap_lower_run_1`).

The round descriptions are read from the series files (pools and bounds). The files do not record why each pool was chosen; the main text states the rule (fix what hits a bound or has $t < 2$ and rerun), which the authors applied by hand to the results of the previous rounds. Because the bounds of rounds 1–5 were tuned to earlier results (for example the lower bound of n_{cap}, Sect. S3.4), the final stage below was designed to remove that dependence.

TABLE S10: Calibration rounds. Runs: independent PSO runs (seeds differ); admissible: no calibrated parameter on a bound and all $t > 2$; best: lowest ammonium-deficient calibration NRMSE (%) among all runs and among admissible runs, with the ammonium-deficient validation NRMSE of that run. Round x: two additional series (initial values of n_{cap} and m_{cap} equal to 1; lower bound of n_{cap} equal to 0.1). Final stage: neutral bounds fixed before the runs (Sect. S3.6); in its rows the columns *Hit a bound* and *Admissible* are defined as in rounds 1–5, and the number of calibrated parameters of the best run is in parentheses. The best admissible run of the final stage is the one of `seventh_run_0`.

Round	Series	Runs	Free parameters	Hit a bound	Admissible	Best (all)	Best adm.	Val. of best adm.
1	3	240	1–12	192	40	4.94 (10)	6.29 (1)	26.6
2	9	45	6–11	44	0	4.95 (11)	—	—
3	6	30	4–6	25	5	5.10 (6)	5.90 (4)	24.8
4	11	117	4–11	77	25	4.95 (10)	5.10 (5)	19.6
5	11	110	4–7	60	40	4.99 (6)	5.10 (5)	19.6
add	2	10	4–4	0	10	5.90 (4)	5.90 (4)	24.8
Rounds 1–5, x	42	552	1–12	398	120	4.94	5.10	19.6
Final: `sixth`	8	44	5–7	13	17	4.98 (6)	4.98 (6)	19.7
Final: `seventh`	9	90	4–8	27	40	4.95 (7)	4.98 (7)	19.7
All	59	686	1–12	438	177	4.94	4.98	19.7

C. Calibration subsets

Of the 552 runs of rounds 1–5, 398 ended with a calibrated parameter on a bound and 34 had no parameter on a bound but a t-value below 2; 120 were admissible. Within a calibration subset, replicate runs with different seeds converge to the same optimum (the range of the calibration NRMSE within a subset is below 0.1 percentage points in every subset of Table S11), so the 120 admissible runs represent only 11 distinct solutions, one per subset. The ten runs of the selected series of the final stage (`seventh_run_0`) all reach the same global NRMSE (0.0510) and the same t-values (smallest 4.3).

TABLE S11: Distinct admissible solutions: for each calibration subset with at least one admissible run, the run with the lowest ammonium-deficient calibration NRMSE (%), with its errors on the other datasets. Runs: admissible runs of that subset, which converge to the same optimum (calibration NRMSE range within the subset below 0.1 percentage points in every subset).

Calibrated parameters	k	Adm. runs	PE_Amm	V_Amm	PE_Norm	V_Norm	$\min t$	$\max \lvert\kappa\rvert$
$\mu_{\max}$, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, n_{cap}	5	15	5.1	19.6	20.9	15.5	4.2	0.94
$\mu_{\max}$, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}	4	30	5.7	22.9	21.5	16.5	18.6	0.65
$\mu_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, n_{cap}	4	35	5.9	24.8	21.5	17.3	5.6	0.92
$Y_{X/C}$	1	5	6.3	26.6	22.0	18.7	29.7	—
m_C	1	5	6.3	24.5	21.8	16.8	10.0	—
n_{cap}	1	5	9.5	18.5	18.6	12.2	12.1	—
$\mu_{\max}$	1	5	9.8	18.0	19.1	12.1	32.0	—
$X_{\max}$	1	5	10.2	23.5	19.8	15.1	26.8	—
t_{lag}	1	5	10.2	17.7	19.5	12.0	10.8	—
$\rho_{N,\max}$	1	5	11.0	16.7	20.5	12.6	10.7	—
Y_{X/CO_2}	1	5	11.0	17.0	20.5	12.4	18.9	—

D. Lower bound of n_{cap}

In rounds 1–5 the five-parameter subset was run with several lower bounds of n_{cap} (all other bounds equal; 85 runs). With a lower bound of 0.5 (60 runs in four series) the estimate stayed on the bound ($n_{cap} = 0.5$), with an ammonium-deficient

calibration NRMSE of 5.16% and a validation NRMSE of 20.3%; with 0.4 (5 runs) it stayed on that bound (5.10% and 19.7%); with 0.35 and with 0.3 (5 and 15 runs) it converged inside the interval to $n_{cap} = 0.378$ (5.10% and 19.6%). The estimate is therefore on a bound only if the bound is above about 0.38. Moving the bound from 0.5 to 0.3 lowers the calibration error by 0.06 percentage points and the ammonium-deficient validation error by 0.6; this is consistent with the weak determination of n_{cap} ($t = 4.3$, $\kappa = 0.94$ with $\mu_{\max}$). Lower bounds of 0.35 or below produced failed objective evaluations (Sect. S1.2). The four-parameter calibration subset without $X_{\max}$ converged to $n_{cap} = 0.78$ in both of its variants (lower bound 0.1 and 0.5; 35 runs). The final stage, with the lower bound 0.3 fixed in advance (the lowest value at which most runs still integrate, Sect. S1.2) and an upper bound of 5, gives $n_{cap} = 0.3793$ inside the interval in all ten runs of `seventh_run_0`.

E. Bounds

The bounds of the calibrated parameters are in Table S1 (*Widest range*: over all 59 series; *Final bounds*: those of `seventh_run_0`, fixed in advance for the final stage, Sect. S3.6). $\mu_{\max}$ in [0.05, 1] h^{-1} contains the baseline value (0.1335) and extends from about 0.4 to about 7.5 times it. $X_{\max}$ in [0.8, 2.5] g/L contains the baseline value (1.1537) and the largest biomass in the data (1.45 g/L, nutrient-replete). $Y_{X/C}$ and Y_{X/CO_2} in [0.05, 1] g/g are wide and non-restrictive: glycerol is 39% carbon by mass (3 C in 92.1 g/mol), so for a biomass with more than 39% carbon a yield on glycerol above 1 g/g would violate carbon conservation; the calibrated values (0.339 and 0.359) are far inside the range and the upper bound was never active in the final stage (it was 5 g/g in earlier rounds). n_{cap} in [0.3, 5]: the lower bound is the lowest value at which most runs still integrate (Sect. S1.2; 0.2 failed in every test), $n_{cap} = 1$ is the logistic case of the baseline and 5 is far above the estimate; the lower bounds of earlier rounds are discussed in Sect. S3.4. The parameters added in the final stage have the bounds $\rho_{N,\max} \in [0.02, 0.22]$ g N/(g X·h) (a factor of 11; the lower bound is about three times the peak nitrogen demand of the model, $\mu q \approx 0.007$ g N/(g X·h), Table S3), $q_{\max} \in [0.096, 0.12]$ g N/g X (the lower bound equals the assumed initial quota $q(0) = 0.096$ g N/g X), $q_{\min} \in [0, 0.05]$ g N/g X, $pH_{opt} \in [2, 12]$, $pH_w \in [1, 7]$ (`sixth_6`) or $[3, 10]$ (`seventh_run_7`), $t_{lag} \in [1, 60]$ (`sixth_2`) or $[1, 45]$ h (`seventh_run_3`), $m_C \in [0, 0.5]$ (`sixth_3`) or $[0, 0.1]$ (`seventh_run_4`) g C/(g X·h) and $m_{cap} \in [0, 15]$ (`sixth_7`) or $[0.01, 15]$ (`seventh_run_2`). No published source gives bounds for these parameters. The authors set them as wide, physically admissible ranges, using the rationale stated above where one exists, and kept them identical for all series of the final stage so that the comparison does not depend on earlier results. A wide range reflects the absence of prior information on a parameter; it is not evidence that any particular value is correct.

F. Final stage: neutral bounds, forward screening and nested ladder

Design. The final stage (134 runs in 17 series: `sixth_1`–`sixth_8`, 44 runs, and `seventh_run_0`–`seventh_run_8`, 90 runs; Table S12) uses the same PSO settings, range-scaled cost and calibration dataset as rounds 1–5, but fixes beforehand the bounds of the five parameters of M5 ($\mu_{\max}$ 0.05–1 h^{-1}, $X_{\max}$ 0.8–2.5 g/L, $Y_{X/C}$ and Y_{X/CO_2} 0.05–1 g/g, n_{cap} 0.3–5), identical in all 17 series, and the assumed values of all other parameters (Table S1, column Initial), with $n_{cap}(0) = 0.8$ and $pH_w = 10$ (pH inhibition is off in practice, Table S3). The bounds of the parameters added to the pool are in Sect. S3.5; those of t_{lag}, m_C, m_{cap} and pH_w were changed slightly between the `sixth` and the `seventh` series (all bounds are in the series files). Three questions were asked.

(i) Does the earlier solution survive neutral bounds? The five-parameter pool was rerun (`sixth_1`, `sixth_5`, `seventh_run_0`; 22 runs). All ten runs of `seventh_run_0` are admissible and reach the same solution (SSE 0.10410, range below 10^{-10}; smallest $t = 4.3$). Relative to the selected set of the five rounds (SSE 0.10411, $pH_w = 7$) the four calibrated parameters $\mu_{\max}$, $X_{\max}$, $Y_{X/C}$ and Y_{X/CO_2} change by at most 0.02% and n_{cap} by 0.4% (0.3777 to 0.3793); the global NRMSE of the four datasets changes by at most 0.1 points (Table S20, M5_fifth against M5). The earlier selection is therefore not an artifact of the hand-tuned bounds or of $pH_w = 7$.

(ii) Does any other parameter earn a place? Forward screening: each of the sensitive or assumed parameters that were fixed (t_{lag}, m_C, m_{cap}, $\rho_{N,\max}$, pH_{opt} with pH_w, $q_{\min}$ with $q_{\max}$) was added to the pool of M5, alone or in pairs (13 extension series; the `sixth` series were stopped after 5 or 6 of the 10 planned runs, enough to see whether runs end on a bound, and the `seventh` series extend the six-parameter model with $\rho_{N,\max}$ and have all ten runs). Only 3 of the 13 extension series have admissible runs: $+\rho_{N,\max}$ (`sixth_4`, `seventh_run_1`) and $+\rho_{N,\max} + q_{\max}$ (`seventh_run_6`). In the other ten no run is admissible: t_{lag}, m_C, m_{cap}, $q_{\min}$ and pH_{opt}/pH_w end on a bound or have $t < 2$ in every run (the smallest t of any run is between 0.00 and 0.43 in each of them, Table S12). The lowest SSE of the stage (0.0978, `seventh_run_3`, $k = 7$, $\min t = 0.40$) is therefore not a usable solution.

(iii) Which of the admissible calibration subsets is worth its parameters? A nested ladder of four calibration subsets of the extended model (M4 to M7, numbered by their calibrated parameters), ten runs each, all admissible and with a single optimum per series: M4 (n_{cap} fixed at 0.8; `seventh_run_8`), M5, M6 (M5 $+\rho_{N,\max}$) and M7 (M6 $+q_{\max}$). Table S13 and Fig. S2 give AIC, AICc, BIC with $k+1$ estimated quantities (the k calibrated parameters and the error variance; Sect. S4.1; the small-sample correction $2(k+1)(k+2)/(n-k-2)$ with $n = 40$ is used for AICc), Akaike weights $\exp(-\Delta/2)/\sum\exp(-\Delta/2)$, nested F-tests and the identifiability of each subset, and Table S14 gives the estimates.

- n_{cap} *is earned.* Removing it (M4) raises SSE by 20% (ΔAIC -7.05 and ΔAICc -6.27 for adding it, ΔBIC -5.36; $F = 8.9$, $p = 0.005$), and the ammonium-deficient and nutrient-replete validation errors are higher (22.9% against 19.6% and 16.5% against 15.5%). M4 has the best conditioning (smallest t 18.6, FIM condition number 219), but its fit relies on the assumed $n_{cap} = 0.8$, so it is the fallback when conditioning matters more than fit.
- $\rho_{N,\max}$ *is not earned.* M6 lowers SSE by 4.7% but is inside the two-unit band of M5 for AIC (Δ =+0.09), AICc (+1.05) and BIC (+1.78), with $F = 1.7$, $p = 0.21$; the ammonium-deficient validation error does not improve (19.7% against 19.6%); the FIM condition number grows by a factor of 12 (3.5×10^4 against 2.9×10^3), the smallest t falls from 4.3 to 3.8 and the half-widths of n_{cap} and $\rho_{N,\max}$ reach 51% and 48%; the largest correlation, that of $\mu_{\max}$ and n_{cap}, is 0.953 (0.943 in M5), above the 0.95 screen.
- $q_{\max}$ *adds nothing.* M7 has ΔAICc +4.15 and ΔBIC +5.42 against M5 ($F = 0.0$, $p = 0.84$ against M6), a condition number 143 times that of M5 and $\rho_{N,\max}$ with a half-width of 70% ($t = 2.8$); $q_{\max}$ itself is estimable (0.09817, $t = 77.8$) and lies inside its bounds, unlike in the rounds 1–5 where it ended on its lower bound $q(0)$.

The Akaike weights (AICc) of M4–M7 are 0.02, 0.57, 0.34 and 0.07 (AIC: 0.01, 0.43, 0.41, 0.15; BIC: 0.04, 0.65, 0.27, 0.04); M5 has the largest weight with all three criteria, but M6 is not excluded by AIC or by AICc. M5 is selected as the calibration subset that earns every parameter it has, with the best balance between fit and identifiability; the choice is a judgement from information criteria, F-tests and the FIM, not a test result. *Caveats.* (a) All these criteria assume independent Gaussian residuals; the residuals of a dynamic model are serially correlated (Table S19), so Δ and p rank the calibration subsets and are not tests, and the two-unit band is a convention. (b) The relative half-width of n_{cap} is 46% in M5 and n_{cap} is strongly correlated with $\mu_{\max}$ ($\kappa = 0.943$): n_{cap} is needed by the fit but weakly determined. (c) The decision rule used only calibration data (fit, information criteria, F-tests, FIM). Validation errors were not used in the selection rule. The validation errors of the ammonium-deficient and nutrient-replete datasets appear in the result tables next to these quantities; they agree with the choice but cannot be read as independent confirmation, and they rest on one validation dataset per condition.

TABLE S12: Series of the final stage. Each adds the listed parameters to the pool of the five-parameter set (M5: $\mu_{\max}$, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, n_{cap}), except `seventh_run_8`, which removes n_{cap}. k: calibrated parameters; runs: finished runs (10 planned; the `sixth` series were stopped after 5 or 6); Bound: runs with a calibrated parameter on a bound; Adm.: runs with no parameter on a bound and all $t > 2$; SSE: best optimiser cost; ΔAICc: against M5 (best runs, $k + 1$ estimated quantities); validation: ammonium-deficient validation NRMSE (%) of the best run; $\min t$ and condition number: worst over the finished runs. All series: same PSO settings, range-scaled cost, ammonium-deficient calibration dataset, $pH_w = 10$ and $n_{cap}(0) = 0.8$.

Series	Added	k	Runs	Bound	Adm.	SSE	ΔAICc	Val.	$\min t$	Cond.
`sixth_1`	–	5	6	0	6	0.1041	+0.0	19.6	4.26	2.9×10^3
`sixth_2`	t_{lag}	6	6	2	0	0.0994	+1.1	20.0	0.43	4.3×10^8
`sixth_3`	m_C	6	5	1	0	0.1041	+3.0	19.6	0.04	3.9×10^3
`sixth_4`	$\rho_{N,\max}$	6	5	0	5	0.0993	+1.0	19.7	3.87	3.4×10^4
`sixth_5`	–	5	6	0	6	0.1041	+0.0	19.6	4.26	2.9×10^3
`sixth_6`	pH_{opt}, pH_w	7	6	3	0	0.1002	+4.6	19.0	0.02	2.0×10^{10}
`sixth_7`	m_{cap}	6	5	2	0	0.1021	+2.2	22.8	0.00	9.4×10^9
`sixth_8`	$q_{\max}$, $q_{\min}$	7	5	5	0	0.1041	+6.1	19.6	0.02	4.5×10^8
`seventh_run_0`	–	5	10	0	10	0.1041	+0.0	19.6	4.26	2.9×10^3
`seventh_run_1`	$\rho_{N,\max}$	6	10	0	10	0.0993	+1.0	19.7	3.84	3.5×10^4
`seventh_run_2`	m_{cap}, $\rho_{N,\max}$	7	10	1	0	0.0991	+4.1	19.7	0.05	2.0×10^{11}
`seventh_run_3`	$\rho_{N,\max}$, t_{lag}	7	10	1	0	0.0978	+3.6	19.9	0.40	6.9×10^8
`seventh_run_4`	m_C, $\rho_{N,\max}$	7	10	10	0	0.0990	+4.1	19.7	0.00	2.2×10^{10}
`seventh_run_5`	$q_{\min}$, $\rho_{N,\max}$	7	10	10	0	0.0988	+4.0	19.4	0.18	1.0×10^9
`seventh_run_6`	$q_{\max}$, $\rho_{N,\max}$	7	10	0	10	0.0991	+4.1	19.7	2.68	4.1×10^5
`seventh_run_7`	pH_{opt}, pH_w, $\rho_{N,\max}$	8	10	5	0	0.0985	+7.3	19.3	0.01	5.0×10^{12}
`seventh_run_8`	$-\,n_{cap}$	4	10	0	10	0.1305	+6.3	22.9	18.61	219

G. Data usage and choice of the parameter set

Optimization. Every PSO run was fitted to the ammonium-deficient calibration dataset (PE_Amm_def) only; no validation dataset and no nutrient-replete dataset entered the objective.

Selection. In rounds 1–5 the selected set was the admissible run with the lowest global NRMSE on PE_Amm_def: among the 120 admissible runs the five-parameter solution (5.10%); the lowest calibration error of any run of those rounds was 4.9% (10 parameters, three on a bound, smallest $t < 0.01$). The final stage (Sect. S3.6) re-examined the choice with bounds fixed in advance: it reproduces the solution and compares it with models with fewer and more parameters by AICc, BIC, F-test and identifiability; the reported set (M5) is the best run of `seventh_run_0`. The validation datasets (V_Normal, V_Amm_def_rep) and the nutrient-replete calibration dataset are test data for the extended model. They were computed for every run to follow the behavior of the model (the result tables list them next to the calibration error). In rounds 1–5 they were not considered in the choice of the subsets or the bounds of any round, nor in the selection, which used only the results

TABLE S13: Nested ladder of the final stage (best run of each series; $n = 40$ pooled range-scaled residuals of the ammonium-deficient calibration dataset, $k+1$ estimated quantities including the error variance). SSE: optimiser cost $\widetilde{RSS}$. AIC, AICc and BIC as in Sect. S4.1; Δ: difference to M5 (positive = worse than M5); w: Akaike weight within the four calibration subsets. F-test: nested F-test of the calibration subset against the previous row (SSE reduction per added parameter against the residual variance of the larger model; independent Gaussian residuals assumed, which these time series do not satisfy, so the p-values rank the models and are not tests). Errors are global NRMSE (%) of the best run; $\min t$ and the condition number of the FIM (largest over the ten runs) refer to the ammonium-deficient calibration dataset.

Quantity	**M4** ($k = 4$)	**M5** ($k = 5$)	**M6** ($k = 6$)	**M7** ($k = 7$)
Series	`seventh_run_8`	`seventh_run_0`	`seventh_run_1`	`seventh_run_6`
Added to M5	removes n_{cap}	–	$\rho_{N,\max}$	$\rho_{N,\max}$, $q_{\max}$
Runs (of 10) on a bound / admissible	0 / 10	0 / 10	0 / 10	0 / 10
SSE (best; range over runs)	0.13053	0.10410	0.09926 (0.09926–0.09933)	0.09914 (0.09914–0.10009)
AIC	-219.00	-226.05	-225.96	-224.01
ΔAIC	+7.05	+0.00	+0.09	+2.05
ΔAICc	+6.27	+0.00	+1.05	+4.15
ΔBIC	+5.36	+0.00	+1.78	+5.42
Akaike weight (AICc / AIC / BIC)	0.02 / 0.01 / 0.04	0.57 / 0.43 / 0.65	0.34 / 0.41 / 0.27	0.07 / 0.15 / 0.04
F-test against previous row (F, p)	–	8.89, 0.005	1.66, 0.21	0.04, 0.84
Amm.-deficient cal. / val. NRMSE (%)	5.71 / 22.89	5.10 / 19.64	4.98 / 19.69	4.98 / 19.70
Nutrient-replete cal. / val. NRMSE (%)	21.48 / 16.54	20.91 / 15.53	20.65 / 15.64	20.70 / 15.55
Smallest t (any run)	18.61	4.26	3.84	2.68
FIM condition number (largest)	219	2.9×10^{3}	3.5×10^{4}	4.1×10^{5}
Largest $\lvert\kappa\rvert$ (best run)	0.654	0.943	0.953	0.952

TABLE S14: Estimates of the calibration subsets of the ladder (best run of each series): value, half-width of the 95% interval as % of the estimate and FIM t-value (ammonium-deficient calibration dataset). Bounds of the final stage: $\mu_{\max}$ 0.05–1 h^{-1}, $X_{\max}$ 0.8–2.5 g/L, $Y_{X/C}$ and Y_{X/CO_2} 0.05–1 g/g, n_{cap} 0.3–5, $\rho_{N,\max}$ 0.02–0.22 g N/(g X·h), $q_{\max}$ 0.096–0.12 g N/g X; no run ended on a bound.

Parameter	**M4**	**M5**	**M6**	**M7**
$\mu_{\max}$	0.1468 (7%, 29.7)	0.118 (9%, 21.0)	0.1275 (14%, 14.4)	0.1277 (16%, 12.2)
$X_{\max}$	1.088 (8%, 24.0)	1.025 (5%, 37.9)	1.035 (6%, 30.4)	1.036 (7%, 29.8)
$Y_{X/C}$	0.3352 (9%, 21.6)	0.3391 (8%, 24.7)	0.3423 (8%, 24.5)	0.3424 (8%, 24.3)
Y_{X/CO_2}	0.3551 (11%, 18.6)	0.3591 (10%, 18.9)	0.3633 (10%, 19.8)	0.3635 (10%, 19.6)
n_{cap}	0.8 (fixed)	0.3793 (46%, 4.3)	0.4802 (51%, 3.9)	0.4825 (55%, 3.6)
$\rho_{N,\max}$	—	—	0.08658 (48%, 4.1)	0.06568 (70%, 2.8)
$q_{\max}$	—	—	—	0.09817 (3%, 77.8)

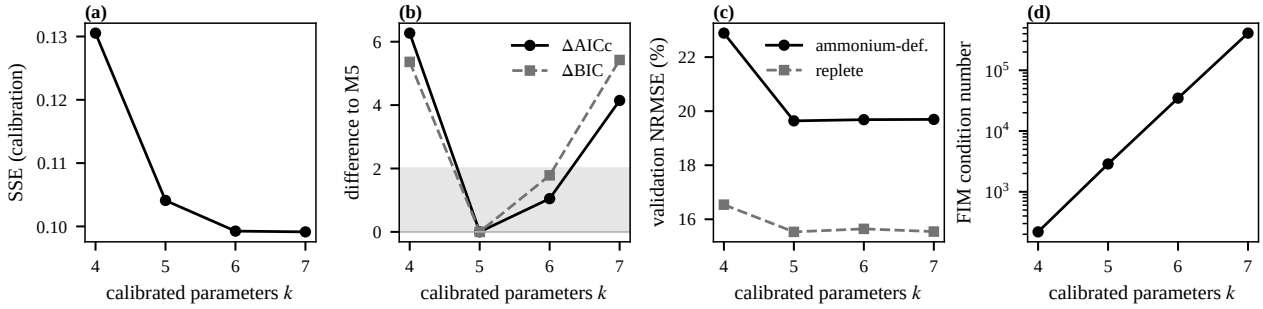


Fig. S2: Nested ladder of the final stage (M4–M7, $k = 4$–7 calibrated parameters; best run of each series): (a) calibration SSE; (b) ΔAICc and ΔBIC against M5 (gray band: two-unit convention); (c) global NRMSE on the two validation datasets (not used in the decision rule, see the text); (d) FIM condition number of the best run (depends on parameter scaling and is only compared within the ladder).

on PE_Amm_def (bound hits, t-values and correlations). In the final stage the decision rule again used only calibration data, but the ladder tables show the validation errors next to the criteria and the authors saw them (Sect. S3.6, caveat (c)).

Calibration error and validation error. Figure S3 shows all 686 runs. Across the 11 distinct admissible solutions of rounds 1–5 the rank correlation between the ammonium-deficient calibration error and the ammonium-deficient validation error is -0.69 ($p = 0.02$, $n = 11$; the replicate runs of one subset are not independent observations), and with the nutrient-replete validation error -0.65 ($p = 0.03$): the five solutions with calibration errors of 6.3% or less validate at 19.6–26.6%, whereas five of the six solutions with calibration errors of 9.5% or more (one-parameter fits, Table S11) validate at 16.7–18.5%, below the selected set. If the runs themselves are used the coefficient is -0.09 ($p = 0.32$, $n = 120$), which is not interpretable

because of the replication. Among the admissible solutions of rounds 1–5 the selected set has the lowest calibration error but not the lowest ammonium-deficient validation error (5.1% and 19.6%, against 11.0% and 16.7% for the single-parameter fit of $\rho_{N,\max}$, Table S11); in the final stage the six- and seven-parameter models reach lower calibration errors (5.0%) with the same validation error (19.7%) and are not selected, for the reasons of Sect. S3.6. The selected set is preferred in this work for fit and identifiability on the calibration data, and with one validation dataset per condition and ten samples per variable the differences among the solutions of Table S11 are indicative only.

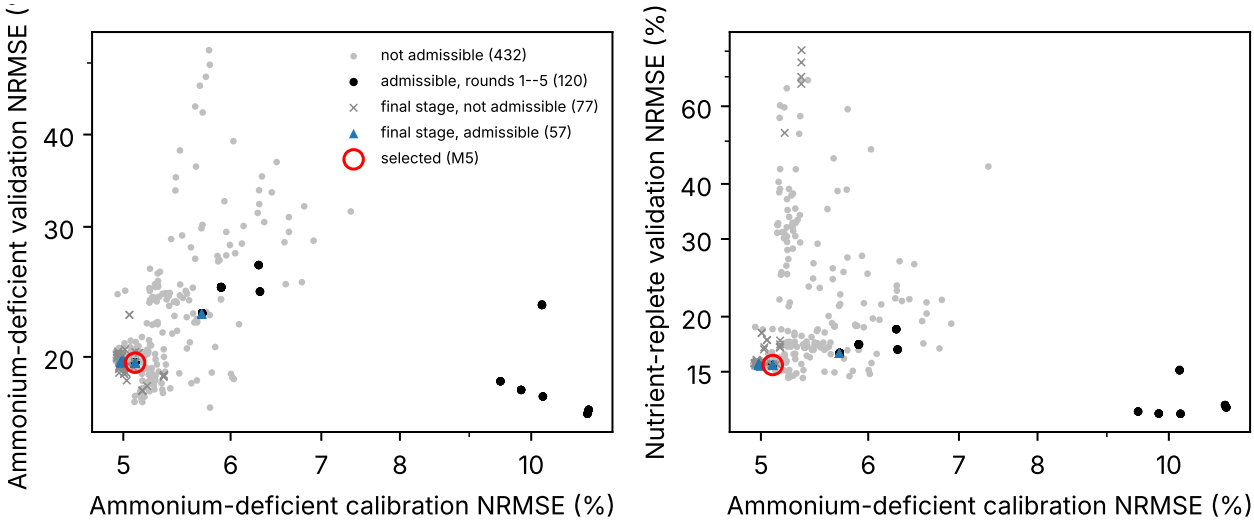


Fig. S3: Ammonium-deficient validation (left) and nutrient-replete validation (right) NRMSE against the ammonium-deficient calibration NRMSE for the 686 runs. Rounds 1–5: black, admissible runs (120, representing 11 distinct solutions: replicates coincide); light gray, not admissible. Final stage: blue triangles, admissible runs; gray crosses, not admissible. Red circle: selected set (M5).

TABLE S15: Series of rounds 1–5 and the two additional series (those of the final stage are in Table S12). Pool: parameters available to the series (each run calibrates a subset of k of them, see k); k: number of calibrated parameters per run; runs; admissible runs; best ammonium-deficient calibration NRMSE (%) over all runs of the series and over its admissible runs. The series of the table were run with the same PSO settings (Sect. S3).

Series	Pool	k	Runs	Adm.	Best all	Best adm.	Round
wide_1	k_O, $\mu_{\max}$, t_{lag}, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, $q_{\min}$, $q_{\max}$, m_C, pH_{opt}, $\rho_{N,\max}$, n_{cap}	11–12	65	0	4.94	—	1
wide_2	k_O, $\mu_{\max}$, t_{lag}, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, $q_{\min}$, $q_{\max}$, m_C, pH_{opt}, $\rho_{N,\max}$, n_{cap}	10	115	0	4.94	—	1
wide_3	k_O, $\mu_{\max}$, t_{lag}, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, $q_{\min}$, $q_{\max}$, m_C, pH_{opt}, $\rho_{N,\max}$, n_{cap}	1	60	40	6.29	6.29	1
second_run_1	$\mu_{\max}$, t_{lag}, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, $q_{\min}$, $q_{\max}$, m_C, pH_{opt}, $\rho_{N,\max}$, n_{cap}	11	5	0	4.95	—	2
second_run_2	k_O, $\mu_{\max}$, t_{lag}, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, $q_{\min}$, $q_{\max}$, m_C, pH_{opt}, n_{cap}	11	5	0	5.01	—	2
second_run_3	$\mu_{\max}$, t_{lag}, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, $q_{\max}$, m_C, pH_{opt}, n_{cap}	9	5	0	4.99	—	2
second_run_4	$\mu_{\max}$, t_{lag}, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, $q_{\max}$, m_C, pH_{opt}, n_{cap}	9	5	0	4.99	—	2
second_run_5	$\mu_{\max}$, t_{lag}, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, $q_{\max}$, pH_{opt}, n_{cap}	8	5	0	4.99	—	2
second_run_6	$\mu_{\max}$, t_{lag}, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, $q_{\max}$, n_{cap}	7	5	0	4.99	—	2
second_run_7	$\mu_{\max}$, t_{lag}, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, n_{cap}	6	5	0	4.99	—	2
second_run_8	$\mu_{\max}$, t_{lag}, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, pH_{opt}, n_{cap}	7	5	0	4.99	—	2
second_run_9	$\mu_{\max}$, t_{lag}, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, pH_{opt}	6	5	0	4.99	—	2
third_run_1	$\mu_{\max}$, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, pH_{opt}, n_{cap}	6	5	0	5.10	—	3
third_run_2	$\mu_{\max}$, $X_{\max}$, $Y_{X/C}$, Y_{X/CO_2}, n_{cap}	5	5	0	5.16	—	3

Series	Pool	k	Runs	Adm.	Best all	Best adm.	Round
third_run_3	$\mu_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	4	5	5	5.90	5.90	3
third_run_4	$\mu_{\max}, Y_{X/C}, Y_{X/CO_2}, q_{\min}, n_{cap}$	5	5	0	5.67	—	3
third_run_5	$\mu_{\max}, Y_{X/C}, Y_{X/CO_2}, q_{\min}, q_{\max}, n_{cap}$	6	5	0	5.67	—	3
third_run_6	$\mu_{\max}, Y_{X/C}, Y_{X/CO_2}, q_{\max}, n_{cap}$	5	5	0	5.90	—	3
fourth_run_1	$\mu_{\max}, t_{lag}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	6	5	0	4.99	—	4
fourth_run_2	$\mu_{\max}, t_{lag}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	5	5	0	5.39	—	4
fourth_run_3	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	5	5	0	5.16	—	4
fourth_run_4	$\mu_{\max}, t_{lag}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, q_{\min}, q_{\max}, m_C, pH_{opt}, n_{cap}, m_{cap}$	10–11	37	0	4.95	—	4
fourth_run_5	$\mu_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	4	20	20	5.90	5.90	4
fourth_run_6	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	5	5	5	5.10	5.10	4
fourth_run_7	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, m_C, n_{cap}$	6	5	0	5.10	—	4
fourth_run_8	$\mu_{\max}, t_{lag}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	6	5	0	4.99	—	4
fourth_run_9	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, q_{\max}, n_{cap}$	6	5	0	5.10	—	4
fourth_run_10	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, pH_{opt}, n_{cap}$	6	5	0	5.09	—	4
fourth_run_11	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	5	20	0	5.16	—	4
fifth_run_1	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	5	30	0	5.16	—	5
fifth_run_2	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}$	4	30	30	5.72	5.72	5
fifth_run_3	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	4–5	10	10	5.10	5.10	5
fifth_run_4	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	5	5	0	5.10	—	5
fifth_run_5	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	5	5	0	5.10	—	5
fifth_run_6	$\mu_{\max}, t_{lag}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, q_{\max}, n_{cap}$	7	5	0	4.99	—	5
fifth_run_7	$\mu_{\max}, t_{lag}, X_{\max}, Y_{X/C}, Y_{X/CO_2}$	5	5	0	4.99	—	5
fifth_run_8	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, q_{\max}$	5	5	0	5.72	—	5
fifth_run_9	$\mu_{\max}, t_{lag}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, q_{\max}$	6	5	0	4.99	—	5
fifth_run_10	$\mu_{\max}, t_{lag}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	6	5	0	4.99	—	5
fifth_run_11	$\mu_{\max}, X_{\max}, Y_{X/C}, Y_{X/CO_2}, q_{\max}, n_{cap}$	6	5	0	5.10	—	5
dif_start_run_1	$\mu_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	4	5	5	5.91	5.91	add
ncap_lower_run_1	$\mu_{\max}, Y_{X/C}, Y_{X/CO_2}, n_{cap}$	4	5	5	5.90	5.90	add

S4. Performance on all datasets

A. Error metrics

Table S16 gives the per-variable and global NRMSE of the main text with four decimals, for the baseline model, the initial set and the selected set on the four datasets; Table S17 gives the errors in native units, as mean absolute percentage error (MAPE) and as mean bias (simulation minus data); Table S18 gives the AIC and BIC. The initial set is the extended model with the base values of the series files before any calibration: it reproduces neither condition (global NRMSE in Table S16), which shows that the improvement of the selected set comes from the calibrated parameters and not from the added structure alone.

TABLE S16: NRMSE of the baseline model, the initial set (base values of the series files, uncalibrated) and the selected set of the extended model (M5), per variable and global, on the four datasets (fractions of the data range, four decimals). The baseline was calibrated on PE_Normal and the selected set on PE_Amm_def; all other rows are tests without refit.

Dataset	Model	X	C	N	pH	Global
PE_Normal	Baseline	0.1300	0.0924	0.2928	0.1601	0.1849
	Initial set	0.1866	0.1282	0.2642	0.2189	0.2055
	Extended	0.1648	0.1772	0.2674	0.2116	0.2091
V_Normal	Baseline	0.0932	0.0937	0.1995	0.0578	0.1231
	Initial set	0.0943	0.0926	0.2036	0.0561	0.1245
	Extended	0.1011	0.1940	0.2074	0.0751	0.1553
PE_Amm_def	Baseline	0.4661	0.4876	0.4770	0.4650	0.4740
	Initial set	0.0799	0.1898	0.0165	0.0759	0.1101
	Extended	0.0495	0.0545	0.0177	0.0683	0.0510
V_Amm_def_rep	Baseline	0.4830	0.4255	0.3430	0.3845	0.4123
	Initial set	0.1669	0.1493	0.0265	0.2481	0.1676
	Extended	0.1269	0.3263	0.0246	0.1765	0.1964

AIC and BIC are secondary to the error reduction: several parameters of the extended model take the values of [1] or were calibrated in some way before this work, so the criteria penalize a number of parameters that does not describe the whole

TABLE S17: RMSE in native units (g/L; g N/L for N; pH units), mean absolute percentage error (MAPE, %) and mean bias (simulation minus data) per variable, for the baseline model and the selected set of the extended model. Samples with a measured value of exactly zero are excluded from the MAPE.

Dataset	**Model**	**Metric**	X	C	N	pH
PE_Normal	Baseline	RMSE	0.1583	0.2132	0.0335	0.1388
		MAPE (%)	11.0	3.9	3.2	1.9
		bias	−0.1103	+0.0096	−0.0110	+0.1023
	Extended	RMSE	0.2007	0.4090	0.0306	0.1834
		MAPE (%)	14.4	8.1	2.9	2.4
		bias	−0.1487	−0.3223	−0.0068	+0.1524
V_Normal	Baseline	RMSE	0.0914	0.2500	0.0253	0.0462
		MAPE (%)	7.1	4.3	2.1	0.6
		bias	+0.0376	−0.0136	+0.0054	−0.0256
	Extended	RMSE	0.0991	0.5178	0.0263	0.0601
		MAPE (%)	8.1	9.9	2.2	0.7
		bias	−0.0424	−0.4200	+0.0147	+0.0415
PE_Amm_def	Baseline	RMSE	0.4106	1.3513	0.0173	0.2232
		MAPE (%)	38.4	28.0	36291.2	2.9
		bias	−0.3238	+1.0967	+0.0144	+0.1872
	Extended	RMSE	0.0436	0.1511	0.0006	0.0328
		MAPE (%)	5.2	2.3	52.3	0.4
		bias	+0.0042	+0.0002	−0.0003	+0.0019
V_Amm_def_rep	Baseline	RMSE	0.4318	1.0027	0.0109	0.2140
		MAPE (%)	41.7	22.6	13262.6	2.9
		bias	−0.3737	+0.8353	+0.0077	+0.1939
	Extended	RMSE	0.1134	0.7688	0.0008	0.0982
		MAPE (%)	10.3	16.4	59.1	1.2
		bias	+0.0511	−0.6614	−0.0003	−0.0787

TABLE S18: Pooled sum of squared range-scaled residuals $\widetilde{RSS}$, AIC and BIC (Gaussian errors with unknown variance, $n = 40$, $m + 1$ estimated quantities; $m = 3$ for the baseline model and $m = 5$ for the extended model). Lower is better; the values have no meaning on their own and are only compared within a dataset. On the datasets where a model was not calibrated they measure predictive agreement and not the cost of a fit.

Dataset	**Model**	$\widetilde{RSS}$	AIC	BIC
PE_Normal	Baseline	1.3682	-127.0	-120.3
	Extended	1.7484	-113.2	-103.1
V_Normal	Baseline	0.6061	-159.6	-152.8
	Extended	0.9652	-137.0	-126.8
PE_Amm_def	Baseline	8.9871	-51.7	-45.0
	Extended	0.1041	-226.1	-215.9
V_Amm_def_rep	Baseline	6.7987	-62.9	-56.1
	Extended	1.5430	-118.2	-108.1

calibration effort (the extended model has 21 parameters, 5 calibrated here, and the baseline has 14, three calibrated). Both criteria are computed on the residuals of each dataset, including datasets on which the model was not calibrated, where they measure predictive agreement and not the cost of a fit.

Formulas. For Gaussian errors with unknown variance, $n = 40$ pooled range-scaled residuals, m calibrated parameters and one unit for the variance, $\mathrm{AIC} = n \ln(\widetilde{RSS}/n) + 2(m + 1)$ [12], $\mathrm{BIC} = n \ln(\widetilde{RSS}/n) + (m + 1) \ln n$ [13] and $\mathrm{AICc} = \mathrm{AIC} + 2(m + 1)(m + 2)/(n - m - 2)$, with $m = 3$ for the baseline model and $m = 5$ for the extended model. Here $\widetilde{RSS}$ is the sum of the squared residuals of X, C, N and pH, each divided by the range of that variable in the dataset. In the nested comparisons, Δ is the value of the larger calibration subset minus that of the smaller one, and the Akaike weight of subset i is $e^{-\Delta_i/2} / \sum_j e^{-\Delta_j/2}$ over the subsets compared. The baseline values are computed in the same way, so both models can be compared on each dataset.

B. Residual diagnostics

Table S19 gives the Durbin–Watson and Anderson–Darling statistics, with the caveats of its caption: ten samples per variable make them indicative only. The residuals of a dynamic model on a time series are serially correlated by construction (a model error persists over several samples), so DW values below 2 are expected; the diagnostic is reported to show where the model errors are systematic (Fig. S4), not as a test of the calibration.

C. Trajectories and values at the sampling times

The trajectories of the four variables of the four datasets are in Fig. 2 of the main text. Fig. S4 gives the range-scaled residuals at the sampling times.

TABLE S19: Residual diagnostics on the range-scaled residuals: Durbin–Watson statistic (DW; near 2: no first-order autocorrelation; the pooled value uses the within-variable differences of the four variables) and Anderson–Darling statistic with small-sample correction (A^{2*}; a normal sample with estimated parameters exceeds 0.752 about 5% of the time). Ten samples per variable make both indicative only, and the pooled residuals mix four variables with different error structure.

Dataset	**Model**	DW X	DW C	DW N	DW pH	DW pooled	A^{2*} pooled
PE_Normal	Baseline	0.53	1.85	1.80	0.18	1.34	0.45
	Extended	0.29	0.39	2.09	0.11	1.00	0.76
V_Normal	Baseline	1.03	1.59	1.66	1.45	1.55	1.44
	Extended	0.62	0.40	1.44	0.79	0.91	0.51
PE_Amm_def	Baseline	0.06	0.09	0.13	0.08	0.09	1.29
	Extended	1.43	2.94	1.12	1.12	1.71	0.83
V_Amm_def_rep	Baseline	0.07	0.10	0.41	0.12	0.15	1.16
	Extended	0.68	0.16	1.79	0.26	0.24	1.71

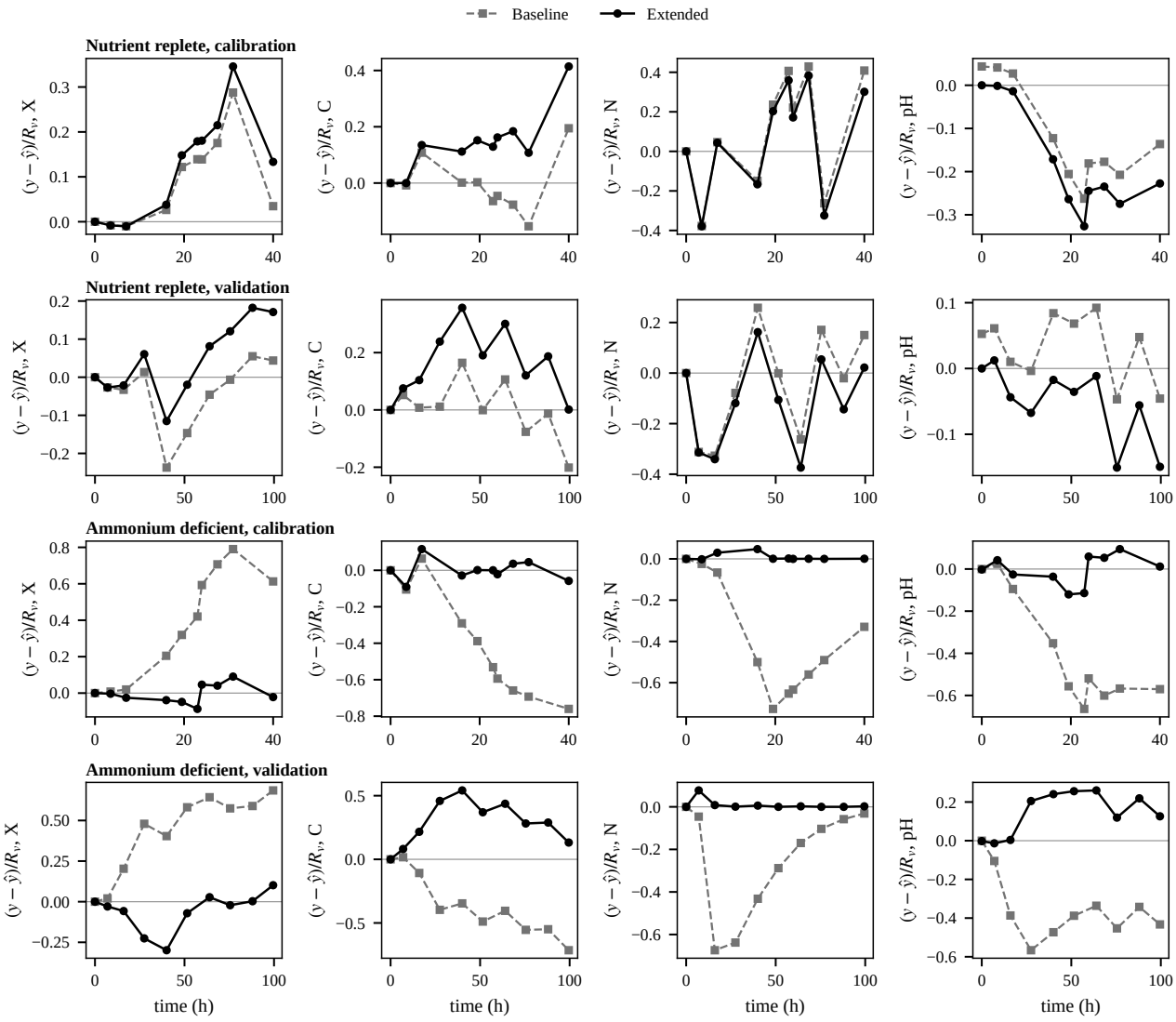


Fig. S4: Range-scaled residuals $(y - \hat{y})/R_v$ of the baseline model (dashed gray, squares) and the selected set of the extended model (solid black, circles) at the sampling times of the four datasets; R_v is the range of the measured variable in the dataset.

S5. Internal behaviour of the extended model

All quantities in this section are model-derived (not measured) and are simulated on the initial states of the validation datasets (Table S5) with the selected set.

TABLE S20: Parameter sets considered besides the selected one (M5, `seventh_run_0`): global NRMSE (%) on the four datasets; M5_fifth: selected set of the five-round analysis (previous analysis, bounds and $pH_w = 7$ differ); L4, L6, L7: the other models of the nested ladder of Sect. S3.6 (best run of `seventh_run_8`, `seventh_run_1`, `seventh_run_6`); k: calibrated parameters; $\min t$: smallest FIM t-value; bounds: calibrated parameters that ended on a bound. The other sets come from runs of the series of Table S10 (Q5–Q8: targeted exploratory runs, Sect. S6). Among the admissible sets, M5 combines a low calibration error with the best identifiability reached by a set that includes n_{cap}; the sets with fewer or more parameters are listed to show the range of results.

Id	Description	k	PE_Amm	V_Amm	PE_Norm	V_Norm	$\min t$	bounds
M5_fifth	Selected set of the five-round analysis (fifth_run_3, pH_w = 7). Selected by the rule: lowest ammonium-deficient calibration NRMSE, no bound hits, all $t > 2$	5	5.1	19.6	20.9	15.5	4.25	none
M5	Extended, 5 parameters (selected). Final stage, series seventh_run_0, run k5_ee6419__r04	5	5.1	19.6	20.9	15.5	4.26	none
M4	Extended, 4 parameters (n_{cap} fixed 0.8). Same without the exponent n_{cap}: best-identified	4	5.7	22.9	21.5	16.5	18.60	none
M4b	Extended, 4 parameters ($X_{\max}$ fixed 1.15). Exponent kept, $X_{\max}$ fixed	4	5.9	24.8	21.5	17.3	5.55	none
QLIM	Droop limitation ON ($q_{\min}$ free). $q_{\min}$ = 0.045 makes N limit growth; n_{cap} at its lower bound, $t(n_{cap})$ = 1.4	5	5.7	22.3	22.1	19.0	1.44	n_{cap}
OVER	Over-parameterised (11 free). Best calibration NRMSE of all runs without bound hits; $\min t = 0.02$	11	4.9	19.8	20.9	15.5	0.02	none
NCAP1	n_{cap} only (1 parameter). Best replete test among the one-parameter fits	1	9.5	18.5	18.6	12.2	12.10	none
RHO1	$\rho_{N,\max}$ only (1 parameter). Best ammonium-deficient validation among the one-parameter fits	1	11.0	16.7	20.5	12.6	10.69	none
BASE	Baseline (Monod, Diaz-Figueroa 2025). Calibrated on the nutrient-replete calibration data ($Y_{X/N}$, $X_{\max}$, $\mu_{\max}$)	3	47.4	41.2	18.5	12.3	4.92	none
Q5	Droop-only plateau (n_{cap} fixed 0.8, $q_{\min}$ free). Exploratory: the plateau is attributed to N limitation alone; $X_{\max}$ runs to its upper bound	5	5.5	19.0	32.4	66.2	1.75	$X_{\max}$
Q6	Selected model plus $q_{\min}$ free. Exploratory: $q_{\min}$ added to the five calibrated parameters of the selected model	6	5.1	19.5	21.1	15.6	0.29	n_{cap}
Q7	Selected model plus $q_{\max}$, $q(0) = q_{\max}$. Exploratory: initial quota tied to the quota ceiling, $q_{\max}$ calibrated ($q_{\max}$ bound 0.03 to 0.2)	6	5.0	19.7	20.3	16.4	3.32	none
Q8	Selected model with constant maintenance. Exploratory: m_{cap} = 0 so that $m(X) = m_C$; the same five parameters recalibrated	5	5.1	22.8	21.0	17.7	4.14	none
L4	Extended, 4 parameters (n_{cap} fixed at 0.8). Final stage, series seventh_run_8, run k4_348cfe__r08	4	5.7	22.9	21.5	16.5	18.61	none
L6	Extended, 6 parameters (selected five + $\rho_{N,\max}$). Final stage, series seventh_run_1, run k6_bbd771__r05	6	5.0	19.7	20.6	15.6	3.87	none
L7	Extended, 7 parameters (selected five + $\rho_{N,\max}$ + $q_{\max}$). Final stage, series seventh_run_6, run k7_56e12f__r03	7	5.0	19.7	20.7	15.5	2.81	none

Nitrogen. In the ammonium-deficient validation initial conditions external nitrogen falls below 10^{-3} g N/L at 17.5 h, after which growth continues on the internal pool: the quota falls from 0.096 to 0.049 g N/g X at 100 h. In the nutrient-replete condition the quota stays between 0.095 and 0.096 g N/g X. The Droop factor f_{Nq} stays at or above 0.9980 in all four datasets, so the limitation f_{Nq} is not binding in the calibrated model (Fig. S5); the plateau is set by the capacity factor f_{cap}, which reaches zero at $X = X_{\max} = 1.025$ g/L at about 30 h in both conditions.

Gases. The simulated dissolved CO_2 ranges from 0.091 to 2.669 g/L (highest in the ammonium-deficient validation condition; the loss term of Eq. (S9) is negligible, Sect. S8, so CO_2 accumulates). Dissolved O_2 starts at 10^{-4} g/L in the validation datasets (Table S5), reaches 90% of saturation within 0.5 h and stays between 6.88e-03 and 7.26e-03 g/L after the first 2 h, so $f_O \geq 0.87$ and oxygen lowers the growth rate by at most 13% in these simulations. CO_2 and O_2 were not measured, so these two trajectories are model outputs only.

Comparison with the baseline model. Unlike the rest of this section, this paragraph uses the ammonium-deficient calibration dataset (0–40 h) and both models at their calibrated sets (baseline: nutrient-replete calibration dataset; extended: ammonium-deficient calibration dataset). The extended model exhausts external nitrogen (below 10^{-3} g N/L) at about 18.5 h; the quota falls from 0.096 to 0.055 g N/g X and the biomass reaches 1.03 g/L at 40 h (measured 1.01 g/L), which is 22.1 g of biomass per g of nitrogen consumed (measured 21.6 g/g). The baseline, whose fixed yield $Y_{X/N} = 10.5$ g/g is equivalent to a constant nitrogen content of 0.095 g N/g X, forms 10.0 g/g, reaches 0.47 g/L at 40 h and leaves 0.012 g N/L in the medium (measured 2.4×10^{-5} g N/L); its biomass is bounded by $X_0 + Y_{X/N}N_0 = 0.60$ g/L. In the nutrient-replete calibration dataset little nitrogen is consumed, both models give the same apparent yield (10.2 g/g) and the quota of the extended model stays between 0.095 and 0.096 g N/g X. Maintenance accounts for 1% of the glycerol demand under ammonium deficiency.

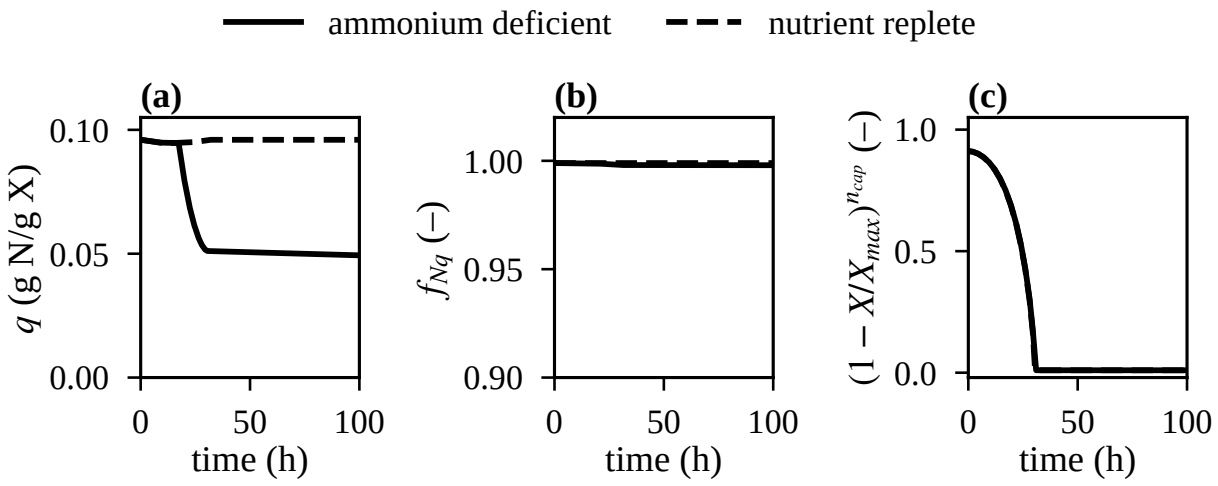


Fig. S5: Internal states of the extended model on the validation initial conditions of both conditions (model-derived, not measured): (a) nitrogen quota q; (b) Droop growth factor f_{Nq}; (c) capacity factor $(1 - X/X_{\max})^{n_{cap}}$, which is the same in both conditions.

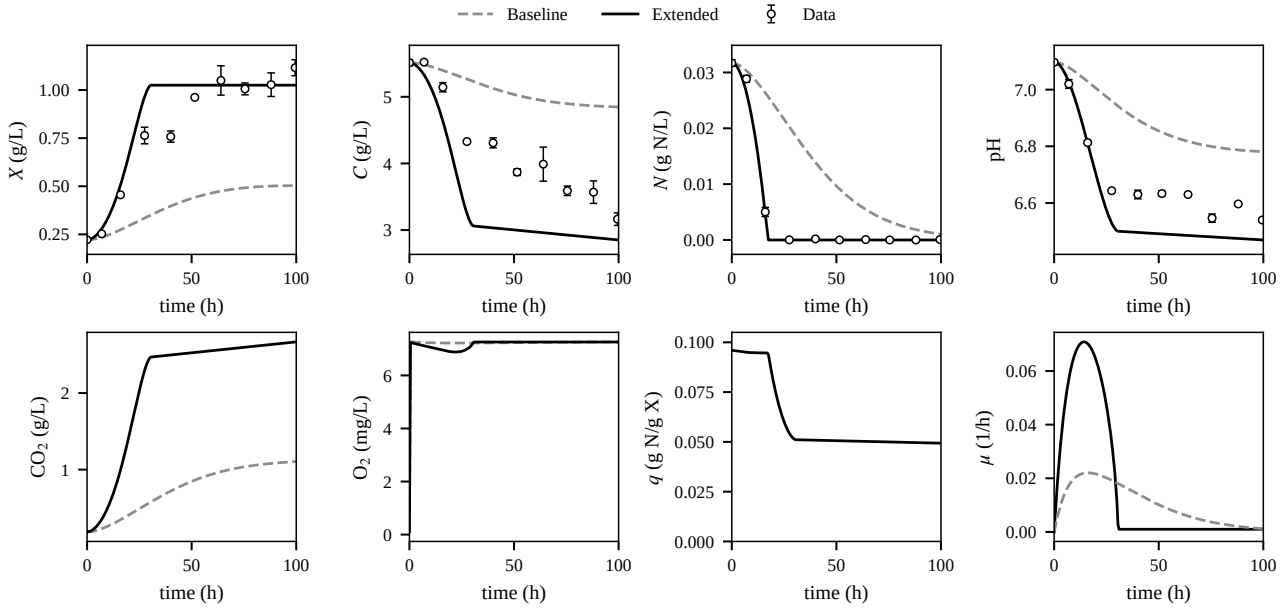


Fig. S6: States of the baseline model (dashed gray) and of the selected set of the extended model (solid black) on the ammonium-deficient validation initial conditions. Dots: data (mean ± SE) for the measured variables. CO_2, O_2, q and μ were not measured.

A. Nitrogen budget

Adding the two balances of Eq. (S8) gives

$$\frac{d}{dt}(N + qX) = -(1 - f_{Nrec})\, k_d\, q\, X, \tag{S17}$$

so the total tracked nitrogen decreases only through the fraction of the quota of dying cells that is not returned to the medium (dead biomass is not tracked). In the ammonium-deficient validation condition the initial tracked nitrogen (0.0317 g N/L external plus 0.096×0.222 g N/L assumed cellular) is 0.0530 g N/L and the simulated value at 100 h is 0.0506 g N/L. Supporting the measured final biomass of 1.12 g/L requires a mean quota of about 0.047 g N/g X at the end, about half the

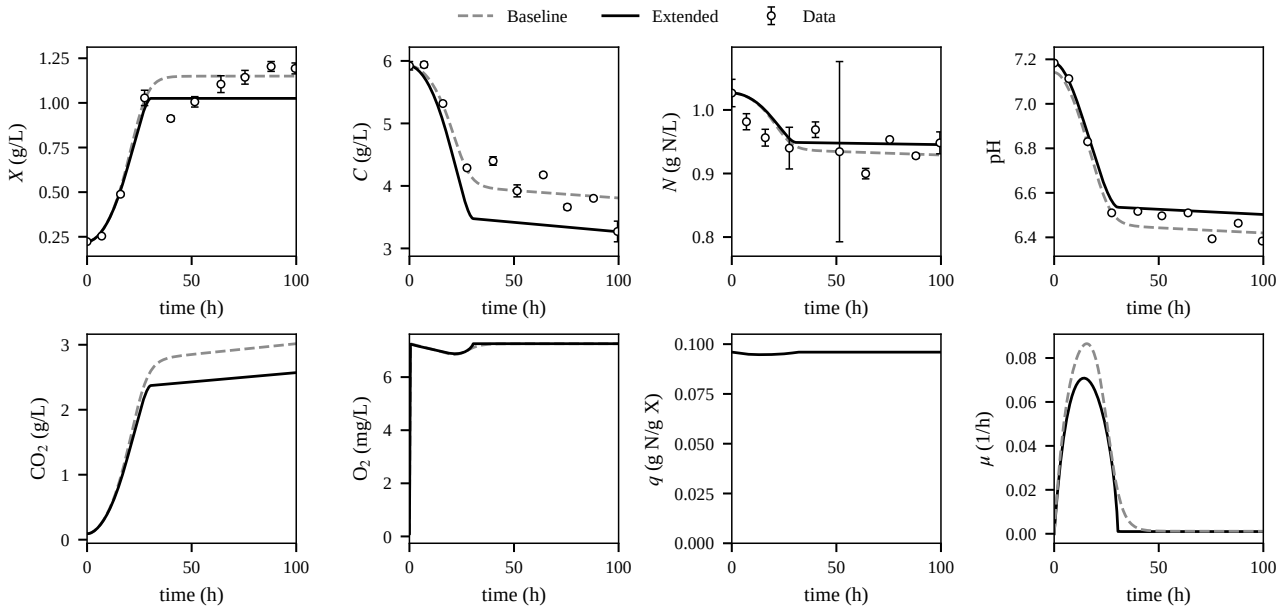


Fig. S7: As Fig. S6, on the nutrient-replete validation initial conditions.

assumed initial value. This is arithmetic on the stored data and on the assumed $q(0)$; it is a mass-balance check and not an independent test: the data cannot distinguish cellular dilution of the quota from nitrogen entering from a source that the model does not include, and the initial quota is assumed.

S6. Ablations and exploratory runs

A. *Ablations of the selected set (no recalibration)*

Table S21 changes one mechanism of the selected set at a time and evaluates the four datasets without recalibration. The question is whether the Droop mechanism does anything in the selected set (A0), where $f_{Nq} \geq 0.9980$ in every dataset and the factor therefore does not reduce growth.

- *A1, limitation on* ($q_{\min} = 0.045$ g N/g X, close to the lowest quota reached): with $q(0) = 0.096$ the factor $f_{Nq} = 1 - q_{\min}/q$ is about 0.53 from the first hour in both conditions, so this change also scales the growth rate under nutrient-replete conditions. The ammonium-deficient calibration error rises to 28.6% and the nutrient-replete errors to 32.7% and 19.8%; the ammonium-deficient validation error falls to 15.4% (from 19.6%). Without recalibration this is a perturbation, not a fit of the Droop limitation (the calibrated version is `QLIM` in Table S20).
- *A2, feedback off* ($q_{\max} = 1$): the uptake no longer stops when the quota is full. The nitrogen NRMSE becomes 42.3% and 29.2% in the ammonium-deficient datasets (global 21.8% and 24.5%) and 535% and 613% in the nutrient-replete ones (global 268% and 307%), where the quota takes up the external nitrogen without a limit.
- *A3, no ammonification* ($f_{Nrec} = 0$): the global NRMSE changes by at most 0.15 percentage points in any dataset.
- *A4, quota frozen* (the quota fixed at its initial value, a constant N:X stoichiometry): the nitrogen NRMSE becomes 65.9% and 124.6% in the ammonium-deficient datasets, with simulated external nitrogen below zero (minimum -0.045 and -0.050 g N/L), while X and C change by less than 0.01 points and pH by at most 0.13 points; the global nutrient-replete errors change by less than 0.1 points.
- *A5, ammonium removed from the charge balance*: the pH error under nutrient-replete conditions changes from 21.2% to 19.4% (calibration) and from 7.5% to 6.5% (validation), and the ammonium-deficient pH changes by 0.01 percentage points or less (6.83 to 6.84% in the calibration dataset). The ammonium term of the charge balance therefore has little effect on the pH in the selected set and does not improve it; it is part of the model for structural consistency with the nitrogen balance, not because it was shown to help.

These ablations support a reading of the quota as a nitrogen pool that makes the nitrogen balance consistent (A2, A4), not as a limiting state (A0 against A1). They do not show that the quota is needed for the growth and glycerol trajectories: X and C are practically identical in A4 and pH differs by at most 0.13 points, because $f_{Nq} \geq 0.9980$ in every dataset, so the factor does not reduce growth.

TABLE S21: Ablations of the selected set (no recalibration): each row changes one mechanism of the extended model and evaluates the four datasets. Global NRMSE (%); N: nitrogen NRMSE (%); $q_{\min}^{traj}$: lowest simulated quota (g N/g X) and $t_{N\,exh}$: time at which external nitrogen falls below 10^{-3} g N/L.

Ablation	**Dataset**	**NRMSE**	N	X_{end}	$q_{\min}^{traj}$	$t_{N\,exh}$
A0 selected	PE_Amm	5.1	1.8	1.025	0.054	18.6
	V_Amm	19.6	2.5	1.025	0.049	17.5
	PE_Rep	20.9	26.7	1.025	0.095	—
	V_Rep	15.5	20.7	1.025	0.095	—
A1 q_min=0.045 (limitation ON)	PE_Amm	28.6	25.1	1.025	0.054	29.5
	V_Amm	15.4	14.5	1.021	0.050	27.5
	PE_Rep	32.7	26.6	1.025	0.096	—
	V_Rep	19.8	26.2	1.025	0.096	—
A2 q_max=1 (uptake feedback OFF)	PE_Amm	21.8	42.3	1.025	0.054	0.5
	V_Amm	24.5	29.2	1.025	0.049	0.4
	PE_Rep	268.1	535.3	1.025	0.096	29.5
	V_Rep	306.5	612.6	1.025	0.096	63.1
A3 f_Nrec=0 (no ammonification)	PE_Amm	5.1	2.0	1.025	0.051	18.5
	V_Amm	19.6	2.6	1.025	0.047	17.4
	PE_Rep	21.0	27.0	1.025	0.095	—
	V_Rep	15.4	20.3	1.025	0.095	—
A4 quota frozen (fixed N:X stoich.)	PE_Amm	33.3	65.9	1.025	0.096	18.4
	V_Amm	65.3	124.6	1.025	0.096	17.2
	PE_Rep	21.0	27.0	1.025	0.096	—
	V_Rep	15.5	20.6	1.025	0.096	—
A5 $NH4^+$ removed from charge balance	PE_Amm	5.1	1.8	1.025	0.054	18.6
	V_Amm	19.6	2.5	1.025	0.049	17.5
	PE_Rep	20.5	26.7	1.025	0.095	—
	V_Rep	15.4	20.7	1.025	0.095	—

B. Targeted exploratory runs

Four targeted calibration series of five runs each were run after round 5 and before the final stage, with the same PSO settings and objective (`explore_Q5` to `explore_Q8`, in a separate results folder; they are exploratory and are not counted among the runs of rounds 1–5 or of the final stage). Each changes one aspect of the selected model of the five rounds, which the final-stage M5 reproduces to within 0.1 points (Sect. S3.6); their results are compared with M5 below.

- *Q5, Droop-only plateau.* $q_{\min}$ calibrated instead of n_{cap} ($n_{cap} = 0.8$ fixed), so that the plateau would have to come from nitrogen limitation: ammonium-deficient calibration 5.5%, validation 19.0%, but $X_{\max}$ ended on its upper bound (2 g/L; $t = 1.75$) and the nutrient-replete errors rose to 32.4% and 66.2% ($q_{\min} = 0.051$ g N/g X). Four of the five runs reached this solution; the fifth ended at 5.7% with $q_{\min}$ on its bound. Not admissible.
- *Q6, $q_{\min}$ added to the selected subset* (six parameters): calibration 5.1%, validation 19.5%, nutrient-replete 21.1% and 15.6%; n_{cap} ended on its lower bound (0.3), and $\mu_{\max}$, n_{cap} and $q_{\min}$ have $t < 2$ (1.0, 0.6 and 0.3). Only one of the five runs reached this solution; the other four ended at 5.2% with $X_{\max}$ and n_{cap} on their bounds and nutrient-replete errors of 40% and 70%. Not admissible.
- *Q7, $q_{\max}$ calibrated with $q(0) = q_{\max}$* (six parameters, $q_{\max} \in [0.03, 0.2]$): calibration 5.0%, validation 19.7%, nutrient-replete 20.3% and 16.4%, with $q_{\max} = 0.085$ ($t = 15.7$), $n_{cap} = 0.478$ ($t = 3.3$), no bound hit and all $t > 3.3$ (admissible); all five runs converge to the same solution. Relative to the selected set (5.1%, 19.6%, 20.9%, 15.5%) the calibration error is 0.1 points lower for one more parameter; the AIC change is $+0.2$ (the extra parameter costs 2, the fit gains 1.8 units), so there is no evidence for it.
- *Q8, constant maintenance* ($m_{cap} = 0$, $m(X) = m_C$; the same five parameters recalibrated): calibration 5.1% (against 5.1%), validation 22.8% (against 19.6%; glycerol 40.0% against 32.6%), nutrient-replete 21.0% and 17.7% (against 20.9% and 15.5%); admissible (smallest $t = 4.1$), five runs converge to the same solution. A constant maintenance gives the same calibration fit and a worse validation, mostly in glycerol; this favors the biomass-dependent form modestly and rests on one validation dataset.

The replicates of Q7, Q8 and of the selected series converge to the same solution, which is consistent with a single optimum inside the bounds for those subsets (it does not prove it); the replicates of Q5 and Q6 do not. The exploratory results are not used to choose the parameter set.

S7. SENSITIVITIES AND IDENTIFIABILITY

A. Relative sensitivities

For parameter θ_j and measured variable $v \in \{X, C, N, pH\}$ the relative sensitivity is

$$S_{v,j} = \frac{1}{n_t} \sum_{i=1}^{n_t} \left| \frac{\theta_j}{\max_i |\hat{y}_{v,i}|} \frac{\partial \hat{y}_{v,i}}{\partial \theta_j} \right|, \tag{S18}$$

where $\partial \hat{y}/\partial \theta_j$ is a central difference with $\pm 5\%$ perturbation of the parameter, evaluated at the sampling times of the ammonium-deficient calibration dataset, with the base and perturbed simulations sharing initial states and time points. Normalizing by $\max_i |\hat{y}_{v,i}|$ (and not by $\hat{y}_{v,i}$) avoids division by values close to zero, such as N after exhaustion. A parameter is non-sensitive if $S_{v,j} < 0.01$ for every variable. At the selected set 9 of the 21 parameters are sensitive and 12 are not (Table S22, Fig. S8). The sensitive parameters that are not calibrated are $q_{\max}$, t_{lag}, k_O and pH_{opt}; a parameter can be sensitive and still not estimable (for example when its effect is collinear with another one), but fixing sensitive parameters at assumed values means that the calibrated values depend on them. The series results bear on this choice. In the seven series (35 runs) that added t_{lag} to a pool of 7 or fewer parameters, none ended with t_{lag} on a bound, but t_{lag} had the smallest t-value of the pool in every run (0.4 to 1.7, with values of 23 to 41 h against 7 h) and a correlation of at least 0.999 with $\mu_{\max}$, with a calibration error of 4.99% (5.39% in one series) against 5.10% for the selected set. $q_{\max}$ ended on a bound in 169 of the 265 runs that calibrated it, including all 20 runs of the four final series that did, always on its lower bound $q(0) = 0.096$ g N/g X (a smaller value starts with the uptake closed); calibrated together with $q(0)$ (Q7, Sect. S6.2) it was estimable ($t = 15.7$) but gave no improvement. k_O ended on its lower bound in 101 of 125 runs (all with 10 to 12 free parameters) and, calibrated alone, had $t = 0.46$ and a calibration error of 9.8% against 11.0% for the initial set. In 247 multi-parameter runs pH_{opt} took values over the whole range of its bounds (2 to 12; median 6.2) with no relation to the calibration error (median 5.2%, rank correlation -0.01), and calibrated alone ($t = 1.58$) it lowered the error from 11.03% to 10.99%. The final stage, with the bounds of Sect. S3.5, points the same way: t_{lag} added to M5 or M6 (16 runs) again had a smallest t of 0.43 and 0.40 in its two series, with 3 runs on a bound; m_C, m_{cap} and pH_{opt} with pH_w ended on a bound or had $t < 2$ in every run of their series (Table S12); and $q_{\max}$, calibrated with $\rho_{N,\max}$ in [0.096, 0.12], lay inside its bounds (0.09817, $t = 77.8$) but did not improve the criteria (ΔAICc $+3.10$ against M6, Table S13). The four parameters were therefore fixed: calibrating them either put them on a bound or left them not estimable from the ammonium-deficient calibration dataset, without improving the fit.

TABLE S22: Relative sensitivities $S_{v,j}$ of the 21 parameters of the extended model at the selected set (ammonium-deficient calibration sampling times; $\pm 5\%$ central differences). Parameters are ordered by their largest value; a parameter is sensitive if one value reaches 0.01.

Parameter	X	C	N	pH	**Calibrated**
$X_{\max}$	0.35	0.18	0.023	0.022	yes
$\mu_{\max}$	0.31	0.16	0.12	0.023	yes
$q_{\max}$	2.7e-04	1.3e-04	0.27	5.7e-05	fixed
$Y_{X/C}$	7.6e-04	0.21	1.0e-04	5.2e-05	yes
t_{lag}	0.12	0.059	0.064	9.1e-03	fixed
n_{cap}	0.08	0.04	0.018	5.7e-03	yes
k_O	0.039	0.019	0.015	2.9e-03	fixed
Y_{X/CO_2}	3.0e-04	1.5e-04	4.7e-05	0.029	yes
pH_{opt}	0.012	6.0e-03	3.2e-03	8.5e-04	fixed
k_d	5.7e-03	2.8e-03	1.6e-03	3.4e-04	fixed (non-sensitive)
k_C	4.7e-03	2.3e-03	1.7e-03	3.4e-04	fixed (non-sensitive)
m_C	1.3e-04	3.6e-03	5.2e-05	1.8e-05	fixed (non-sensitive)
$\rho_{N,\max}$	6.2e-06	3.1e-06	3.0e-03	5.8e-07	fixed (non-sensitive)
m_{cap}	9.0e-05	2.9e-03	2.4e-05	1.4e-05	fixed (non-sensitive)
Y_{X/O_2}	1.4e-03	6.8e-04	3.6e-04	9.8e-05	fixed (non-sensitive)
f_{Nrec}	8.8e-07	4.4e-07	1.0e-03	1.9e-07	fixed (non-sensitive)
pH_w	5.6e-04	2.8e-04	9.6e-05	3.9e-05	fixed (non-sensitive)
$q_{\min}$	3.4e-04	1.7e-04	1.2e-04	2.5e-05	fixed (non-sensitive)
$Y^m_{O_2}$	1.1e-04	5.8e-05	4.8e-05	8.6e-06	fixed (non-sensitive)
$Y^m_{CO_2}$	3.8e-07	1.9e-07	8.0e-08	2.7e-05	fixed (non-sensitive)
k_{Nup}	5.5e-06	2.8e-06	2.3e-05	3.7e-07	fixed (non-sensitive)

B. Fisher information matrix

The FIM is $\mathbf{F} = \sum_v \mathbf{G}_v^\top \mathbf{G}_v / \sigma_v^2$ with $\sigma_v^2 = \sum_i e_{v,i}^2/(n_t - 1)$ from the calibration residuals, and $\mathbf{G}_v$ the sensitivity matrix of variable v built with central differences of relative step 10^{-3}. Standard errors and t-values come from the pseudo-inverse of the sub-matrix of the calibrated parameters, with 95% intervals $\theta_j \pm 1.96\,\mathrm{SE}_j$; the correlation of two estimates is $\kappa_{jk} = C_{jk}/\sqrt{C_{jj}C_{kk}}$ and the threshold for high correlation is $|\kappa| > 0.95$. Table S23 gives the estimates of the selected set

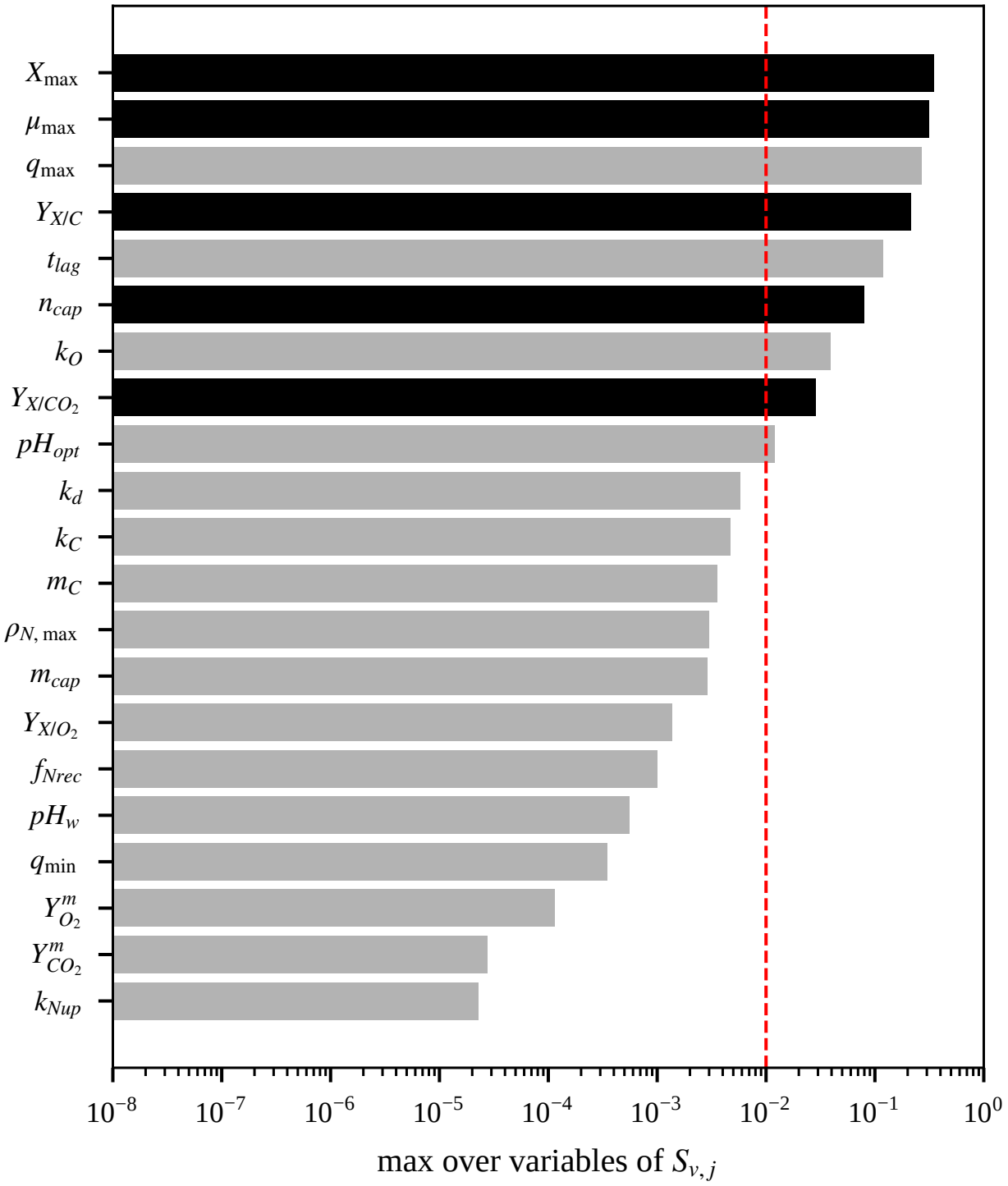


Fig. S8: Largest relative sensitivity of each parameter over the four measured variables at the selected set. Black: calibrated; gray: fixed. Dashed red line: threshold 0.01.

and Table S24 the correlations; the condition number of $\mathbf{F}$ is 2.9×10^3 (it depends on parameter scaling and is not comparable between models). The weakest parameter is n_{cap} ($t = 4.3$; the half-width of its 95% interval is 46% of the estimate), strongly correlated with $\mu_{\max}$ ($\kappa = 0.94$); κ is below 0.95 for every pair. The baseline model has $t = 4.9$ ($Y_{X/N}$), 17.3 ($X_{\max}$) and 11.8

($\mu_{\max}$), with a largest correlation of 0.76. The thresholds $t > 2$ and $|\kappa| > 0.95$ are pragmatic screens, not universal standards: $t > 2$ is equivalent to a 95% interval that excludes zero, and the same two screens, with the same thresholds, are applied to candidate parameterizations in [9], which builds on the iterative reparameterization method HIPPO [10]; the sequential fixing of Sect. S3.2 follows the same idea, applied by hand.

TABLE S23: Identifiability of the calibrated parameters of the extended model (selected set, FIM at the ammonium-deficient calibration dataset): estimate, standard error (SE), 95% interval, relative interval size ($100 \times 1.96\,\mathrm{SE}/\theta$ is the half-width in %) and $t = \theta/\mathrm{SE}$. Baseline model: $t = 4.9$ ($Y_{X/N}$), 17.3 ($X_{\max}$) and 11.8 ($\mu_{\max}$).

Parameter	θ	SE	95% interval	rel. size (%)	t
$\mu_{\max}$	0.118	0.00561	[0.107, 0.129]	9	21.0
$X_{\max}$	1.025	0.027	[0.972, 1.08]	5	37.9
$Y_{X/C}$	0.3391	0.0137	[0.312, 0.366]	8	24.7
Y_{X/CO_2}	0.3591	0.019	[0.322, 0.396]	10	18.9
n_{cap}	0.3793	0.089	[0.205, 0.554]	46	4.3

TABLE S24: Correlation κ_{jk} between the estimates of the calibrated parameters of the extended model (selected set). The largest $|\kappa|$ is that of $\mu_{\max}$ and n_{cap} (0.94, just below the 0.95 screen).

	$\mu_{\max}$	$X_{\max}$	$Y_{X/C}$	Y_{X/CO_2}	n_{cap}
$\mu_{\max}$	1.00	0.01	−0.20	−0.14	0.94
$X_{\max}$	0.01	1.00	0.50	0.32	0.11
$Y_{X/C}$	−0.20	0.50	1.00	0.27	−0.23
Y_{X/CO_2}	−0.14	0.32	0.27	1.00	−0.18
n_{cap}	0.94	0.11	−0.23	−0.18	1.00

All these quantities are linearizations at one parameter point. A significant t-value is necessary but not sufficient for practical identifiability, and a non-significant one does not prove that a parameter has no effect on the data, only that its effect cannot be separated from that of the other calibrated parameters within these ten observations per variable.

S8. Package conventions that affect interpretation

1) The CO_2 loss term uses k_{a7} as a first-order rate constant and is negligible over the cultures (about 4.5×10^{-7} h^{-1} at $pK_{a7} = 6.35$), so the simulated CO_2 accumulates (Sect. S5).
2) Phosphate, citrate and CO_2 enter the charge balance in g/L, and the ammonium term in mol/L. The ammonium term is therefore weighted about 44 to 192 times less than on a common molar basis (molar masses of CO_2 44, KH_2PO_4 136 and citric acid 192 g/mol). Nitrogen assimilation thus acidifies the medium in the model, but weakly (ablation A5); the buffers, Y_{X/CO_2} and pH_{alk} are lumped scaling terms, and the simulated pH is not interpreted as a mechanistic prediction of the buffer chemistry.
3) The initial CO_2 of each dataset is back-calculated from the measured initial pH through the constraint (without the ammonium term for the ammonium-deficient datasets; Table S5).
4) The capacity and maintenance factors are implemented with a maximum with zero (Sect. S1.2); the baseline model keeps its original form.
5) The cost function, the sensitivity window and the FIM mode are options of the package (`cost_scaling`, `sens_window`, `fim_mode`); the settings used for the reported results are those of Sects. S3 and S7.
6) The trajectories, internal states, errors and sensitivities of this Supplementary Material are produced from the final results (CasADi/IDAS); the values at the sampling times are computed by simulating at those times.

Data and code availability

The experimental workbook, the package, the calibration outputs and the scripts that generate the tables and figures of this supplement are in the project repository [11].

Use of AI-assisted tools

The authors developed the model and the calibration framework. The AI assistant Claude (Anthropic) was used to identify caveats in the analysis, to write and run scripts that analyse the calibration outputs in batch instead of one by one (sensitivities, identifiability, ablations and the exploratory runs of Sect. S6.2), and to draft and edit the text and to produce the figures and tables of this Supplementary Material and of the main paper. The authors reviewed and edited the content and are responsible for it.

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